

Citation: BIAN Haotian, ZHUANG Ruixue, WANG Jiabao, BIAN Yufang, LIANG Lichan. Parental Burnout and Adolescents' Internet Gaming Addiction: Roles of Basic Psychological Need Satisfaction, Effortful Control and Gender, *Psychological Development and Education*, 2026 (1).

Parental Burnout and Adolescents' Internet Gaming Addiction: Roles of Basic Psychological Need Satisfaction, Effortful Control and Gender

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Abstract To examine the influence of parental burnout on adolescents' Internet gaming addiction and mechanism, the Basic Psychological Need Satisfaction Scale, Internet Gaming Addiction Scale, and Early Adolescent Temperament Questionnaire were applied to investigate adolescents, and the Parental Burnout Assessment was employed to investigate fathers and mothers. Ultimately, 2669 families' triadic data were collected. After controlling for the subjective family socioeconomic status perceived by adolescents, the results showed that (1) parental burnout significantly and positively predicted adolescents' Internet gaming addiction; (2) parental burnout increased the risk of adolescents' Internet gaming addiction by diminishing their need satisfaction for relatedness, autonomy, and competence. When comparing the differences in the mediating effects, only the mediating effect of autonomy need satisfaction was significantly larger than that of competence need satisfaction; (3) the latter part of the three mediating effects of “parental burnout → basic psychological need satisfaction(relatedness, autonomy, competence) → Internet gaming addiction” were moderated by adolescents' effortful control and gender. Among lower effortful control and male individuals, relatedness need satisfaction, autonomy need satisfaction, and competence need satisfaction all had stronger impacts on Internet gaming addiction. These findings provided insights for the prevention and intervention of adolescents' Internet gaming addiction.

DOI 10.16187/j.cnki.issn1001-4918.2026.01.11-en

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Keywords parental burnout, Internet gaming addiction, basic psychological need satisfaction, effortful control, adolescents

1 Introduction

The 2021 National Report on Internet Usage Among Minors, published by the China Internet Network Information Center (2022), reports that the proportion of minors who frequently play games online in China is 62.3%. Appropriate online gaming can meet individual entertainment and social needs, promote socialization, and enhance cognitive and emotional abilities (Granic et al., 2014; Green & Bavelier, 2003; Przybylski, 2014; Wu et al., 2015). However, excessive online gaming poses a potential risk of addiction. Internet gaming addiction refers to the difficulty individuals have in controlling their excessive or compulsive use of online games, resulting in physical and mental health issues or functional impairments (Ma et al., 2017). Adolescents are a vulnerable group when it comes to Internet gaming addiction (Li et al., 2016), and such addiction can negatively impact various aspects of their physical and mental health, as well as their academic development. This includes severe sleep problems, increased hostility, more depressive moods, and poor academic performance (Borgonovi, 2016; Brunborg et al., 2014; Chiu et al., 2004; Hamre et al., 2022; Tereshchenko et al., 2021). Therefore, it is of significant practical importance to explore the causes of Internet gaming addiction among adolescents for promoting their healthy development.

According to the ecosystem theory, the family is the microenvironment for individual development, and parents, as key figures in a child's life, have a crucial impact on the psychological and social competence development (Bronfenbrenner & Ceci, 1994). Previous studies have indicated that a lack of parental warmth, psychological distress, and neglectful parenting may lead to Internet gaming addiction among adolescents (Lin et al., 2021; Peng & Zhou, 2007; Yin & Xian, 2021). A study focusing on adolescents with Internet addiction has also found that their dissatisfaction with family and parents is significantly higher than that of their peers (Lam et al., 2009). Therefore, it is essential to consider

the potential impact of parenting-related factors on adolescent Internet gaming addiction. Studies by Chen et al. (2022) and Yuan et al. (2022) have demonstrated that parental burnout is an important predictor of various problem behaviors in adolescents, but little is known about the relationship between parental burnout and adolescent Internet gaming addiction. Therefore, building on previous research, the present study focuses on the impact mechanism of parental burnout on adolescent Internet gaming addiction. This includes examining the mediating mechanisms of how parental burnout influences adolescent Internet gaming addiction, as well as the moderating mechanisms through which the negative effects of parental burnout on adolescents are exacerbated or mitigated. An in-depth exploration of these issues will provide theoretical guidance and empirical support for preventing and alleviating adolescent Internet gaming addiction.

1.1 Relationship between parental burnout and adolescent Internet gaming addiction

Parental burnout results from enduring exposure to chronic parenting stress (Roskam et al., 2017; Cheng et al., 2021). It is characterized by overwhelming exhaustion related to one's parental role, loss of parental enjoyment, emotional distancing from children, and contrast between the previous and current parental self (Roskam et al., 2018). Firstly, parents experiencing burnout face emotional exhaustion in parenting, and it is difficult for them to maintain a positive state for raising their children. Adolescence is a sensitive period for emotional experiences, and when teenagers do not perceive emotional warmth from their parents, they may turn to online gaming to fulfill their emotional needs, leading to gaming addiction. Eisenberg et al. (2001) reported that the negative emotions expressed by parents during parent-child interactions can affect children's social understanding, potentially resulting in behavioral problems. Secondly, parents suffering from burnout may feel weary of their parenting roles, deliberately disconnect from their children, and experience a decline in their sense of parenting efficacy. This limits their proactive parenting. Previous studies have indicated that parents with low proactive parenting may neglect to guide and discipline their children's behaviors in advance

and express less affection, resulting in various behavioral issues in children (Pettit & Bates, 1989). Moreover, parents with burnout may even exhibit neglectful or violent parenting behaviors (Mikolajczak et al., 2018), which can drive adolescents to immerse themselves in virtual worlds as a way to cope with the adverse effects of negative parenting (Chapman et al., 2006). Empirical evidence has shown that adolescents who experience parental neglect or harsh parenting are likely to develop Internet gaming addiction (Lin et al., 2021; Yu et al., 2023). Therefore, this study proposes Hypothesis 1: Parental burnout significantly positively predicts adolescent Internet gaming addiction, suggesting that the more severe parental burnout denotes a higher level of adolescent Internet gaming addiction.

1.2 Role of basic psychological need satisfaction in the relationship between parental burnout and adolescent Internet gaming addiction

Further exploration of the specific mechanisms through which parental burnout affects adolescent Internet gaming addiction can facilitate the prevention and mitigation of such addiction among adolescents. Basic psychological need satisfaction represents the intrinsic motivation behind individual behavior (Ryan & Deci, 2000) and includes three dimensions: autonomy need satisfaction (the desire to make one's own choices), relatedness need satisfaction (the desire for love and support from those around), and competence need satisfaction (the desire to be competent for certain tasks). When psychological needs are met, individuals typically develop toward self-improvement and self-enhancement; conversely, unmet needs may cause psychological and behavioral issues. Existing studies have demonstrated that basic psychological need satisfaction mediates the impact of parental psychological distress on adolescent Internet gaming addiction (Yin & Xian, 2021). On this basis, it can be inferred that basic psychological need satisfaction may mediate the relationship between parental burnout and adolescent Internet gaming addiction.

Specifically, parental burnout can lead to impaired parent-child relationships (Gillis & Roskam, 2019). When adolescents cannot perceive a

close connection with their parents, it undermines their relatedness need satisfaction (Gao et al., 2022). Positive parent-child relationships are a source of adolescents' sense of autonomy, and poor relationships can threaten their autonomy need satisfaction (Karabanova & Poskrebysheva, 2013). Negative parent-child relationships may also hinder the development of adolescents' sense of competence, impacting their competence need satisfaction (Sim, 2000). The needs-satisfaction theory states that when psychological needs are suppressed, neglected, displaced, or only superficially or indirectly met, it may induce addiction (Suler, 1999). Online gaming allows for interaction among players, which satisfies the relatedness needs of adolescents. Players can freely choose characters, purchase equipment, and use skills, thus satisfying their autonomy needs. Competing with other players to achieve higher levels in the game further fulfills their competence needs (Wang et al., 2023). Taken together, this study proposes Hypothesis 2: The three dimensions of basic psychological need satisfaction mediate the relationship between parental burnout and adolescent Internet gaming addiction.

However, the influence of different dimensions of basic psychological need satisfaction may vary (Xu et al., 2021). Parent-child interaction is the primary way to establish and maintain emotional connections between parents and children. Parents experiencing burnout are less likely to engage in the interaction, which may cause a deficiency in adolescents' relatedness need satisfaction. Additionally, as a social component, online gaming is appealing for adolescents with unmet relatedness needs and who seek social support within these games. On this basis, this study proposes Hypothesis 3: The mediating effect of relatedness need satisfaction in the relationship between parental burnout and adolescent Internet gaming addiction is the strongest among the three dimensions.

1.3 Moderating role of effortful control and gender differences

Effortful control refers to the ability to inhibit a dominant response to perform a subdominant response and/or activate subdominant responses as a form of self-regulation (Rothbart & Bates, 2006). The theory of action

control posits that motivation and willpower jointly determine individual behavior (Kuhl, 1984). Motivation helps individuals choose goal-directed behaviors that maximally value their needs, while willpower enables them to overcome distractions and engage in corresponding goal-directed behaviors (Zhang & Jin, 2019). High effortful control serves as a protective factor for healthy individual development; for instance, Li et al. (2010) pointed out that adolescents with high effortful control typically exhibit fewer problematic behaviors regarding Internet usage. Those with high effortful control are also likely to adopt adaptive coping strategies such as cognitive restructuring and problem-solving when facing negative life events, rather than defaulting to passive responses (Carver & Connor-Smith, 2010). Luo et al. (2017) suggested that effortful control can buffer the negative effects of accumulated ecological risks on adolescent Internet addiction. It can be inferred that adolescents with high effortful control may reassess the risks and benefits of gaming when their different basic psychological needs are unmet, thereby suppressing their excessive cravings for gaming to avoid addiction. In contrast, adolescents with low effortful control may seek immediate gratification, and in situations where their basic psychological needs are unmet, they find it easier to be attracted to the positive emotional experiences of online gaming, resulting in addiction. Prior studies have also found that self-control moderates the relationship between escapism and problematic smartphone use, indicating that poor self-control ability increases the likelihood of problematic smartphone use due to escapism (Li et al., 2021). Therefore, this study proposes Hypothesis 4: Adolescents' effortful control can moderate the effects of the three dimensions of basic psychological need satisfaction on Internet gaming addiction.

Gender is also a key influencing factor in adolescent Internet gaming addiction. It has been found that the rate of Internet gaming addiction is higher among boys than among their female peers (Stavropoulos et al., 2013; Toker & Baturay, 2016; Lin et al., 2021). This may be due to boys having more experience with gaming and stronger motivations to play than girls (Ko et al., 2005). Additionally, Stavropoulos et al. (2013) noted that boys experience a higher level of immersive engagement in online gaming

than girls. This strong sense of immersion satisfies individuals’ diverse basic psychological needs (Fang et al., 2019) and may cause a higher risk of gaming addiction among boys. Therefore, this study proposes Hypothesis 5: There are gender differences in the impact of the three dimensions of basic psychological need satisfaction on adolescent Internet gaming addiction.

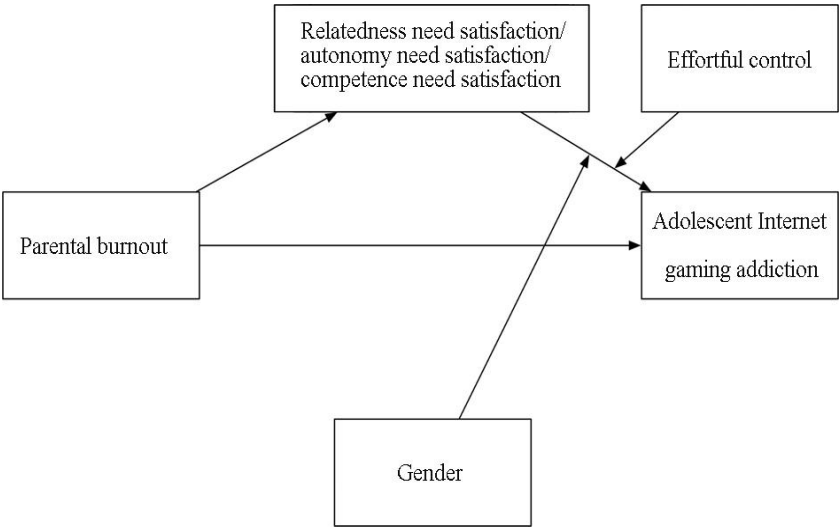


Fig. 1 Hypothesis model to be tested

2 Study method

2.1 Participants

This study employed a cluster sampling method among Grade 7 students and their parents from 13 schools in Beijing and Hainan Province. The researchers were graduate students majoring in psychology who had undergone standardized training. Before completing the questionnaires, informed consent was obtained from all participants, who were informed of their freedom to complete any questionnaires of interest and their right to withdraw at any time. A total of 3,274 sets of family data were collected, and strict data inclusion criteria were applied. Data from 605 families were excluded due to more than 30% missing responses on any single variable or invalid answers, resulting in 2,669 sets of valid data, yielding a valid response rate of 81.52%. Among the 2,669 student participants, 1,418 were

boys (53.10%), and 1,251 were girls (46.90%). In subsequent analyses, male participants were coded as 0 and female participants as 1.

2.2 Instruments

2.2.1 Parental burnout

The Parental Burnout Assessment developed by Roskam et al. (2018) was used, with the Chinese version translated by Cheng et al. (2020). The scale consists of 23 items, which cover four dimensions: exhaustion in one's parental role, contrast with previous parental self, feelings of being fed up with one's parental role, and emotional distancing from one's children. A 7-point scoring system was employed, ranging from 0 ("never") to 6 ("every day"). The assessment was administered separately to fathers and mothers, and the average score was calculated; a higher score indicates greater parental burnout. In this study, the Cronbach's α coefficients for the paternal and maternal burnout scales were 0.98. The average scores were combined to create a composite measure of parental burnout.

2.2.2 Basic psychological need satisfaction

The Basic Psychological Need Satisfaction and Frustration Scale, developed by Chen et al. (2015), was adopted in this study. This scale consists of 24 items, covering three dimensions: relatedness need satisfaction, autonomy need satisfaction, and competence need satisfaction. The scale was rated on a 5-point scoring system, ranging from 1 ("not at all true") to 5 ("very true"). Reverse scoring was applied to negatively worded items, and the average score of all items was calculated; a higher score indicates a greater degree of adolescents' basic psychological need satisfaction. In this study, the Cronbach's α coefficient for the relatedness, autonomy, and competence need satisfaction subscales was 0.82, 0.73, and 0.84, respectively.

2.2.3 Internet gaming addiction

The Internet Gaming Addiction Scale, adapted from the DSM-IV Criteria for Pathological Gambling by Gentile (2009), was employed in this study. The participants were asked to report the frequency of symptoms related to Internet gaming addiction in this scale, with 11 items in total. A 3-point scoring system was used, where 1 denotes "never," 2 "sometimes," and 3 "frequently." Responses were recoded as follows: no

point for “never,” 0.5 points for “sometimes,” and 1 point for “frequently.” The total score was then calculated, with a higher score meaning a greater level of Internet gaming addiction. In this study, the Cronbach’s α coefficient for this scale was 0.88.

2.2.4 Effortful control

Effortful control was measured using three subdimensions—activation control, attention, and inhibitory control—from the Early Adolescent Temperament Questionnaire (Capaldi & Rothbart, 1992). The questionnaire comprises 15 items. It was rated on a 5-point scoring system, ranging from 1 (“strongly disagree”) to 5 (“strongly agree”). Negatively worded items were reverse-scored, and the average score for all items was calculated; a higher score means greater effortful control. In this study, the Cronbach’s α coefficient for this scale was 0.82.

2.2.5 Control variables

In line with previous findings (Wang et al., 2022), subjective family socioeconomic status was included as a control variable in all analyses. It was measured using the 10-rung MacArthur Scale (Adler et al., 2000). Higher steps correspond to higher scores, suggesting that adolescents perceive their family members as having higher levels of education and social status, and that the quality of family life is better.

2.3 Data processing

SPSS 26.0 was used to conduct descriptive statistics and correlation analyses on the data. Based on these analyses, the Process macro was employed for moderated mediation analysis. The Bootstrap method was used to obtain standard errors and confidence intervals for the mediating effect estimates, with the number of resamples set to 5,000. The significance of the parameters was evaluated based on the estimated 95% confidence intervals.

3 Results

3.1 Test for common method bias

Except for parental burnout, which was reported by parents, all other variables were obtained from adolescents, suggesting that the

adolescent-reported results may have common method bias. Then, Harman's single-factor method was employed to test the results reported by adolescents, and eight principal components were extracted. The first factor accounted for 25.46% of the total variance, which is below the 40% critical threshold proposed by Zhou and Long (2004). Therefore, common method bias was not a significant concern in this study.

3.2 Correlation among variables

Correlation analyses were conducted on parental burnout, relatedness need satisfaction, autonomy need satisfaction, competence need satisfaction, effortful control, Internet gaming addiction, adolescent gender, and adolescents' perceived subjective family socioeconomic status. It was found that parental burnout was significantly negatively correlated with subjective family socioeconomic status, the three dimensions of basic psychological need satisfaction, and effortful control, while it was significantly positively correlated with Internet gaming addiction. All three dimensions of basic psychological need satisfaction had a significant positive correlation with subjective family socioeconomic status and effortful control, and a significant negative correlation with Internet gaming addiction. Effortful control was significantly positively associated with subjective family socioeconomic status and significantly negatively associated with Internet gaming addiction. There was a significant negative association between Internet gaming addiction and subjective family socioeconomic status. Moreover, adolescent gender was significantly related to relatedness need satisfaction, competence need satisfaction, effortful control, and Internet gaming addiction. Specifically, girls had higher levels of relatedness need satisfaction and effortful control, while boys had higher levels of competence need satisfaction and Internet gaming addiction (Table 1).

3.3 Parental burnout and adolescent Internet gaming addiction: the roles of basic psychological need satisfaction, effortful control, and gender

The maximum variance inflation factor (VIF) among the variables in this study was 2.69, which is below the critical value of 4, suggesting no

severe multicollinearity issue. All variables, except for gender, were standardized. Following the recommendations of Wen and Ye (2014), the moderated mediating effects were tested by a stepwise approach.

Table 1 Descriptive statistics and correlation analysis of variables

	1	2	3	4	5	6	7	8
1 Gender	1							
2 SES	—	1						
3 Parental burnout	−0.01	−0.13 ***	1					
4 Relatedness need satisfaction	0.05 *	0.26 ***	−0.20 ***	1				
5 Autonomy need satisfaction	0.03	0.22 ***	−0.17 ***	0.71 ***	1			
6 Competence need satisfaction	−0.09 **	0.27 ***	−0.14 ***	0.68 ***	0.71 ***	1		
7 Effortful control	0.05 **	0.27 ***	−0.22 ***	0.64 ***	0.65 ***	0.64 ***	1	
8 Internet gaming addiction	−0.23 ***	−0.15 ***	0.17 ***	−0.33 ***	−0.36 ***	−0.33 ***	−0.49 ***	1
<i>M</i> ± <i>SD</i> _(overall)	—	5.96 ± 1.56	1.71 ± 0.97	3.75 ± 0.76	3.53 ± 0.68	3.46 ± 0.80	3.65 ± 0.60	2.11 ± 2.08
<i>M</i> ± <i>SD</i> _{boys}	—	5.98 ± 1.57	1.72 ± 0.96	3.71 ± 0.73	3.51 ± 0.66	3.53 ± 0.77	3.62 ± 0.61	2.56 ± 2.21
<i>M</i> ± <i>SD</i> _{girls}	—	5.95 ± 1.54	1.70 ± 0.97	3.79 ± 0.79	3.54 ± 0.70	3.38 ± 0.84	3.68 ± 0.59	1.61 ± 1.81

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (the same as below).

In the first step, a mediation model was established using Model 4 in PROCESS to test whether the three dimensions of basic psychological need satisfaction ^① mediate the relationship between parental burnout and adolescent Internet gaming addiction, while controlling for subjective family socioeconomic status in all analyses. The results indicated that both the total effect and the direct effect of parental burnout on adolescent Internet gaming addiction were significant ($\beta = 0.15, p < 0.001$; $\beta = 0.10, p < 0.001$), supporting Hypothesis 1. Parental burnout predicted all three dimensions of basic psychological need satisfaction ($\beta = -0.17, p < 0.05$; $\beta = -0.15, p < 0.001$; $\beta = -0.10, p < 0.001$). Moreover, the predictive effects of the three dimensions of basic psychological need satisfaction on adolescent Internet gaming addiction were also significant ($\beta = -0.09, p < 0.05$; $\beta = -0.19, p < 0.001$; $\beta = -0.12, p < 0.001$). Mediation analyses revealed that all three dimensions of basic psychological need satisfaction mediated the effect of parental burnout on adolescent Internet gaming addiction, and the mediating effects accounted for 9.66%, 17.63%, and

^① The findings on relatedness, autonomy, and competence need satisfaction are presented in that order in the following sections.

7.58%, respectively, thus supporting Hypothesis 2. Further differential testing demonstrated that the mediating effect of autonomy need satisfaction was significantly greater than that of competence need satisfaction, whereas the difference in the mediating effect between autonomy and relatedness need satisfaction was not significant. Additionally, there was no significant difference between the mediating effects of relatedness and competence need satisfaction, which does not support Hypothesis 3 (Table 2).

Table 2 Mediating effects of the three dimensions of basic psychological need satisfaction and their differences

Variables	β	95% bootLLCI	95% bootULCI
Relatedness need satisfaction	0.015 *	0.006	0.025
Autonomy need satisfaction	0.027 *	0.018	0.038
Competence need satisfaction	0.012 *	0.006	0.019
Relatedness need satisfaction v.s. autonomy need satisfaction	-0.012	-0.023	0.003
Relatedness need satisfaction v.s. competence need satisfaction	0.003	-0.009	0.015
Autonomy need satisfaction v.s. competence need satisfaction	0.015 *	0.003	0.029

In the second step, three moderated mediation models were established. Model 16 in PROCESS was used to test whether the mediating effects of parental burnout on adolescent Internet gaming addiction through relatedness need satisfaction, autonomy need satisfaction, and competence need satisfaction were moderated by adolescents' effortful control and gender. The results showed that in all three models, the predictive effects of parental burnout on the three dimensions of basic psychological need satisfaction were significant; the predictive effects of the three dimensions of basic psychological need satisfaction on adolescent Internet gaming addiction were significant; the predictive effect of adolescents' effortful control on adolescent Internet gaming addiction was significant; the interaction effects between the three dimensions of basic psychological need satisfaction and effortful control significantly predicted adolescent Internet gaming addiction, indicating that effortful control moderated the relationship between each of the three dimensions of basic psychological need satisfaction and Internet gaming addiction. Therefore, Hypothesis 4 was supported. The predictive effect of gender on adolescent

Internet gaming addiction was significant; the interaction effects between the three dimensions of basic psychological need satisfaction and gender significantly predicted Internet gaming addiction, indicating that gender differences existed in the relationship between each of the three dimensions of basic psychological need satisfaction and Internet gaming addiction. Therefore, Hypothesis 5 was supported (see Table 3).

Table 3 Moderating effects of effortful control and gender

Variables	Equation 1					Equation 2				
	β	<i>t</i>	95% LLCI	95% ULCI	R^2	β	<i>t</i>	95% LLCI	95% ULCI	R^2
Dependent variable: relatedness need satisfaction					0.09	Dependent variable: Internet gaming addiction				
SES	0.24 ***	12.72	0.20	0.27		-0.02	-0.92	-0.05	0.02	
Parental burnout	-0.17 ***	-9.00	-0.20	-0.13		0.06 ***	3.71	0.03	0.10	
Relatedness need satisfaction						-0.08 **	-2.78	-0.13	-0.02	
Effortful control						-0.45 ***	-21.04	-0.50	-0.41	
Relatedness need satisfaction $\times$ effortful control						0.07 ***	4.14	0.03	0.10	
Gender						-0.40 ***	-12.29	-0.47	-0.34	
Relatedness need satisfaction $\times$ gender						0.10 **	2.94	0.03	0.16	
Dependent variable: autonomy need satisfaction					0.07	Dependent variable: Internet gaming addiction				
SES	0.20 ***	10.40	0.16	0.23		-0.02	-0.10	-0.05	0.02	
Parental burnout	-0.15 ***	-7.75	-0.18	-0.11		0.06 ***	3.61	0.03	0.09	
Autonomy need satisfaction						-0.15 ***	-5.40	-0.21	-0.10	
Effortful control						-0.42 ***	-19.02	-0.46	-0.38	
Autonomy need satisfaction $\times$ effortful control						0.08 ***	5.07	0.05	0.11	
Gender						-0.40 ***	-12.39	-0.47	-0.34	
Autonomy need satisfaction $\times$ gender						0.11 ***	3.47	0.05	0.18	
Dependent variable: competence need satisfaction					0.08	Dependent variable: Internet gaming addiction				
SES	0.25 ***	13.55	0.22	0.29		-0.01	-0.59	-0.04	0.02	
Parental burnout	-0.10 ***	-5.44	-0.14	-0.07		0.07 ***	3.93	0.03	0.10	
Competence need satisfaction						-0.17 ***	-5.97	-0.23	-0.11	
Effortful control						-0.41 ***	-18.79	-0.45	-0.37	
Competence need satisfaction $\times$ effortful control						0.08 ***	5.15	0.05	0.11	
Gender						-0.43 ***	-12.89	-0.49	-0.36	
Competence need satisfaction $\times$ gender						0.13 ***	4.12	0.07	0.20	

To better understand the moderation effects, simple slope analyses were conducted. The results found that for boys with low effortful control ($M-1$ SD), all three dimensions of basic psychological need satisfaction significantly predicted Internet gaming addiction ($\beta = -0.14$, $p < 0.001$; $\beta = -0.23$, $p < 0.001$; $\beta = -0.25$, $p < 0.001$). For boys with high effortful control ($M+1$ SD), the predictive effect of relatedness need satisfaction on Internet gaming addiction was not significant ($\beta = -0.01$, $p > 0.05$), while autonomy need satisfaction and competence need satisfaction continued to show significant predictive effects ($\beta = -0.07$, $p < 0.05$; $\beta = -0.09$, $p < 0.01$), albeit at lower levels than for boys with low effortful control. For girls with low effortful control ($M-1$ SD), the predictive effect of relatedness

need satisfaction on Internet gaming addiction was not significant ($\beta = -0.05, p > 0.05$), while autonomy need satisfaction and competence need satisfaction significantly predicted Internet gaming addiction ($\beta = -0.12, p < 0.001$; $\beta = -0.12, p < 0.001$). In contrast, for girls with high effortful control ($M+1\ SD$), relatedness need satisfaction positively predicted Internet gaming addiction ($\beta = 0.08, p < 0.01$), while autonomy need satisfaction and competence need satisfaction had insignificant predictive effects ($\beta = 0.04, p > 0.05$; $\beta = 0.05, p > 0.05$) (Fig. 2).

Additionally, the mediating effects of the three dimensions of basic psychological need satisfaction were moderated by adolescents' effortful control and gender (Table 4).

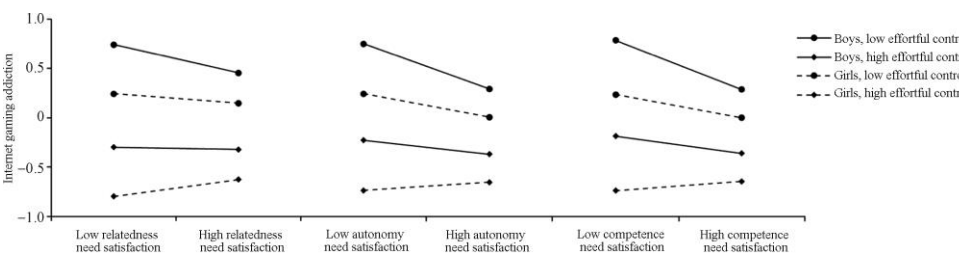


Fig.2 Moderating effects of effortful control and gender on the relationship between the three dimensions of basic psychological need satisfaction and Internet gaming addiction

Table 4 Differences in mediating effects of the three dimensions of basic psychological need satisfaction moderated by effortful control and gender

Mediator variables	Moderator variables	Differences in mediating effects	95% bootLLCI	95% bootULCI
Relatedness need satisfaction	Effortful control	0.011 *	0.005	0.018
	Gender	0.016 *	0.005	0.028
Autonomy need satisfaction	Effortful control	0.012 *	0.007	0.018
	Gender	0.017 *	0.008	0.027
Competence need satisfaction	Effortful control	0.008 *	0.005	0.013
	Gender	0.014 *	0.007	0.022

4 Discussion

This study explores the impact of parental burnout on adolescent Internet gaming addiction and the specific underlying mechanisms involved. It further analyzes the differences in the mediating roles of the three different dimensions of basic psychological need satisfaction. The

findings indicate that parental burnout influences Internet gaming addiction through these three dimensions, and that there are variations in the mediating effects of these dimensions. The study also finds that when relatedness, autonomy, and competence needs are unmet, adolescents with low effortful control, particularly boys, exhibit high levels of Internet gaming addiction.

4.1 Impact of parental burnout on adolescent Internet gaming addiction

This study reveals, for the first time, the impact of parental burnout on adolescent Internet gaming addiction, which expands the understanding of how parental burnout affects children's physical and mental health development. This is consistent with the findings in the field of parenting stress that high levels of parental stress are associated with severe Internet addiction among adolescents (Bozoglan & Kumar, 2022). Parental burnout may lead to adolescent Internet gaming addiction for several reasons. (1) As a family risk factor, parental burnout threatens the quality of parent-child relationships (Gillis & Roskam, 2019). If adolescents feel that the emotional connection with their parents is weakening, they may misinterpret the reward mechanisms and positive emotional experiences in games as a form of solace (Chen et al., 2015) and are highly likely to develop Internet addiction (Hsieh et al., 2018). (2) Parents experiencing burnout may reduce their investment in parenting, leading to a diminished awareness of their children's problematic behaviors (McDaniel & Radesky, 2018), and this lack of parental supervision contributes to adolescents' addiction to Internet gaming (Xu et al., 2012).

4.2 Role of basic psychological need satisfaction in the relationship between parental burnout and adolescent Internet gaming addiction

It is found that high parental burnout is associated with low relatedness, autonomy, and competence need satisfaction in adolescents, which in turn increases the risk of Internet gaming addiction. This finding aligns with the "needs-satisfaction" theory (Suler, 1999). Parents experiencing burnout may reduce their emotional investment in their

children, providing only instrumental support while avoiding excessive depletion of their psychological resources, thus alleviating their own burnout (Mikolajczak et al., 2020). This may lead parents to overlook signals from adolescents seeking satisfaction of their psychological needs, which adversely affects the healthy development of the parent-child relationship and threatens adolescents' basic psychological need satisfaction (Zhu & Jiang, 2022). When individuals' basic psychological needs are unmet, they may seek alternative forms of satisfaction (Ryan & Deci, 2000). Gaming offers anonymity and convenience and also allows adolescents to experience a sense of mastery and achievement, representing a significant means of seeking alternative satisfaction (Kardefelt-Winther, 2014). The greater disparity between inadequate basic psychological needs in the real world and their satisfaction in the virtual world will lead to a higher risk of Internet gaming addiction (Allen & Anderson, 2018). Bender and Gentile (2020) also obtained similar conclusions, indicating that higher need satisfaction in gaming fosters addictive behaviors. Studies in the field of substance addiction have pointed out that when adolescents' basic psychological needs are unmet, there is a significant increase in their behaviors associated with addiction to tobacco, alcohol, and drugs (Ye et al., 2013; Xia & Ye, 2014).

It is important to note that relatedness need satisfaction does not show the strongest mediating role, which is inconsistent with the hypothesis. This may be due to the fact that as children enter adolescence, they seek to satisfy their relatedness needs through interactions with peers (Gao et al., 2022). Conversely, the satisfaction derived from parent-child interactions tends to decrease with age. Nonetheless, this study still reveals differences in the mediating effects of the three dimensions of basic psychological need satisfaction; specifically, the mediating effect of autonomy need satisfaction is significantly greater than that of competence need satisfaction. This may be because parents affected by burnout provide low autonomy-supportive parenting (Yang et al., 2021), leading to inadequate fulfillment of adolescents' autonomy needs. Establishing a sense of identity is a primary developmental task faced by individuals during adolescence (Erikson, 1956), and they are therefore eager to prioritize a sense of

autonomy to meet their developmental needs. Thus, compared to other needs, autonomy needs are more significantly influenced by parental burnout, thereby increasing the risk of Internet gaming addiction.

4.3 Moderating role of effortful control and gender differences

This study examines individual differences in the impact of different dimensions of basic psychological need satisfaction on Internet gaming addiction among adolescents with varying levels of effortful control, from the perspective of the theory of action control. The findings indicate that effortful control plays a “helpful” role when the three dimensions of basic psychological needs are unmet, which aligns with the results from Wang et al. (2017). Specifically, as adolescents’ effortful control abilities increase, there is a gradual decrease in predictive effects of the three dimensions of basic psychological need satisfaction on Internet gaming addiction. When adolescents possess high effortful control, the mediating effects of the three dimensions on the relationship between parental burnout and adolescent Internet gaming addiction are mitigated. This phenomenon can be explained by the psychological decompensation hypothesis (Gao & Chen, 2006), which suggests that adolescents with high effortful control are likely to engage in “constructive compensation,” such as moderate gaming, whereas those with low effortful control tend to exhibit “pathological compensation,” leading to Internet gaming addiction. In other words, adolescents with high effortful control likely have excellent self-regulation skills and can manage their use of Internet gaming to satisfy their basic psychological needs in an effective way, while those with low effortful control may become excessively immersed in gaming and struggle to disengage.

This study also analyzes gender differences in the relationship between the different dimensions of basic psychological need satisfaction and Internet gaming addiction. While both boys and girls are impacted by the three dimensions of basic psychological need satisfaction in Internet gaming addiction, the effect is stronger among boys. This may be because boys generally have a more positive and engaged attitude toward Internet

technology (Schumacher & Morahan-Martin, 2001) and possess more experience in gaming (Ko et al., 2005). Also, boys tend to seek out dominant activities or content online, and interactive video games that emphasize challenge and control have great appeal for them (Chou et al., 2005). Additionally, boys usually experience stronger immersion in these games (Stavropoulos et al., 2013). The immersive experience of gaming meets their basic psychological needs effectively, which further causes a high risk of Internet gaming addiction. Previous studies have also found similar patterns, suggesting that gender significantly moderates the influence of belongingness needs on social media addiction (Chen et al., 2019). In this study, this gender difference is expanded to the relationship between different dimensions of basic psychological need satisfaction and Internet gaming addiction.

Notably, for girls with high effortful control, high satisfaction of relatedness needs is associated with an increased risk of gaming addiction. This may result from two interrelated factors. First, for individuals with low effortful control, the overall Internet gaming addiction is relatively high, which may obscure the impact of relatedness need satisfaction on gaming addiction. In contrast, for those with high effortful control, the overall gaming addiction is relatively low, and the influence of relatedness need satisfaction on gaming addiction is more apparent. Second, girls often demonstrate high social media addiction and relational needs (Varchetta et al., 2020). As such, they may be inclined to fulfil their relatedness needs through Internet gaming, which facilitates gaming addiction. Therefore, it is crucial to be aware of the potential for girls with high effortful control to experience increased risks of gaming addiction due to excessive fulfillment of their relatedness needs.

4.4 Implications

Unlike previous studies that focused solely on the mediating role of overall basic psychological need satisfaction or the moderating effects of effortful control, this study explores the mechanisms underlying different dimensions of basic psychological need satisfaction and also integrates the “needs-satisfaction” theory and the theory of action control within a single

model framework. This integration aids in a deep understanding of how different dimensions of basic psychological need satisfaction and effortful control jointly influence individual behavior. The study identifies gender as a predictor variable rather than merely a control variable, and reveals gender differences in the influence of different dimensions of basic psychological need satisfaction on Internet gaming addiction. Furthermore, it examines the moderating effects of both adolescents' effortful control and gender further, clarifying the varying conditions under which mediating effects occur and their effect sizes. These findings provide insights for preventing and mitigating adolescent Internet gaming addiction. First, adolescence is a turbulent stage for individual physical and mental development, where the desire for independence emerges, yet adolescents struggle to fully separate from their parents. During this time of period, emotional support and behavioral restrictions from parents remain crucial, as they can prevent adolescents from going astray. Second, focusing on and enhancing basic psychological need satisfaction among adolescents, especially autonomy need satisfaction, can significantly reduce their dependence on Internet gaming. To reduce the risk of gaming addiction stemming from the pursuit of psychological needs in virtual worlds, it is crucial to cultivate adolescents' effortful control abilities. Moreover, boys, particularly those with low effortful control, face a high risk of gaming addiction due to unmet basic psychological needs, and they should be given special attention. Finally, when parents find their parenting resources inadequate for the demands they face, they should promptly seek external support to enhance adolescents' health and well-being.

4.5 Limitations and future directions

Several limitations can be addressed in the future. First, due to the cross-sectional design, it is not possible to make causal inferences. Therefore, a longitudinal design is needed in the future to validate and extend the conclusions drawn from this study. Second, while adolescent Internet gaming addiction is treated as the outcome variable in this study, it is important to note that such addiction may also lead to decreased

parental efficacy and subsequent feelings of burnout. Future studies should examine the reciprocal relationship between these two factors using a longitudinal approach. Third, the specific role of relatedness need satisfaction in Internet gaming addiction among girls with high effortful control should be investigated across a broad age range. Finally, besides parents, peers are significant others during adolescence (Brown et al., 1993), and positive peer relationships can provide sufficient emotional support, satisfy basic psychological needs, and reduce the risk of Internet gaming addiction (Sun et al., 2020). In the future, efforts can be made to further analyze the buffering effects of peer support and develop diverse strategies to protect adolescents from the impacts of parental burnout.

5 Conclusion

In summary, the following main conclusions have been obtained. (1) Parental burnout can significantly and positively predict adolescent Internet gaming addiction. (2) Parental burnout affects Internet gaming addiction by impairing the satisfaction of adolescents' relatedness, autonomy, and competence needs. Among the comparisons of mediating effects, only autonomy need satisfaction is found to exert a significantly greater mediating effect than competence need satisfaction. (3) The mediating effects of relatedness, autonomy, and competence need satisfaction are stronger in individuals with low effortful control and males than in high effortful control and females.

References

- Adler, N. E., Epel, E. S., Castellazzo, G., & Ickovics, J. R. (2000). Relationship of subjective and objective social status with psychological and physiological functioning: Preliminary data in healthy white women. *Health Psychology, 19*(6), 586–592.
- Allen, J. J., & Anderson, C. A. (2018). Satisfaction and frustration of basic psychological needs in the real world and in video games predict Internet gaming disorder scores and well-being. *Computers in Human Behavior, 84*, 220–229.

Bender, P. K., & Gentile, D. A. (2020). Internet gaming disorder: Relations between needs satisfaction in-game and in life in general. *Psychology of Popular Media*, 9(2), 266–278.

Borgonovi, F. (2016). Video gaming and gender differences in digital and printed reading performance among 15-year-olds students in 26 countries. *Journal of Adolescence*, 48(1), 45–61.

Bozoglan, B., & Kumar, S. (2022). Parenting styles, parenting stress and hours spent online as predictors of child Internet addiction among children with autism. *Journal of Autism and Developmental Disorders*, 52(10), 4375–4383.

Bronfenbrenner, U., & Ceci, S. J. (1994). Nature-nuture reconceptualized in developmental perspective: A bioecological model. *Psychological Review*, 101(4), 568–586.

Brown, B. B., Mounts, N., Lamborn, S. D., & Steinberg, L. (1993). Parenting practices and peer group affiliation in adolescence. *Child Development*, 64(2), 467–482.

Brunborg, G. S., Mentzoni, R. A., & Frøyland, L. R. (2014). Is video gaming, or video game addiction, associated with depression, academic achievement, heavy episodic drinking, or conduct problems? *Journal of Behavioral Addictions*, 3(1), 27–32.

Capaldi, D. M., & Rothbart, M. K. (1992). Development and validation of an early adolescent temperament measure. *The Journal of Early Adolescence*, 12(2), 153–173.

Carver, C. S., & Connor-Smith, J. (2010). Personality and coping. *Annual Review of Psychology*, 61(1), 679–704.

Chapman, A. L., Gratz, K. L., & Brown, M. Z. (2006). Solving the puzzle of deliberate self-harm: The experiential avoidance model. *Behaviour Research and Therapy*, 44(3), 371–394.

Chen, B., Vansteenkiste, M., Beyers, W., Boone, L., Deci, E. L., Van der Kaap-Deeder, J., ... Verstuyf, J. (2015). Basic psychological need satisfaction, need frustration, and need strength across four cultures. *Motivation and Emotion*, 39(2), 216–236.

Chen, B.-B., Qu, Y., Yang, B., & Chen, X. (2022). Chinese mothers' parental burnout and adolescents' internalizing and externalizing

problems: The mediating role of maternal hostility. *Developmental Psychology*, 58(4), 768–777.

Cheng, H., Wang, W., Wang, S., Li, Y., Liu, X., & Li, Y. (2020). Validation of a Chinese version of the parental burnout assessment. *Frontiers in Psychology*, 11, Article 321.<https://doi.org/10.3389/fpsyg.2020.00321>

Chiu, S. I., Lee, J. Z., & Huang, D. H. (2004). Video game addiction in children and teenagers in Taiwan. *Cyberpsychology & Behavior*, 7(5), 571–581.

Chou, C., Condron, L., & Belland, J. C. (2005). A review of the research on Internet addiction. *Educational Psychology Review*, 17(4), 363–388.

Eisenberg, N., Gershoff, E. T., Fabes, R. A., Shepard, S. A., Cumberland, A. J., Losoya, S. H., ... Murphy, B. C. (2001). Mother's emotional expressivity and children's behavior problems and social competence: Mediation through children's regulation. *Developmental Psychology*, 37(4), 475–490.

Erikson, E. H. (1956). The problem of ego identity. *Journal of the American Psychoanalytic Association*, 4(1), 56–121.

Fang, J., Tang, L., Yang, J., & Peng, M. (2019). Social interaction in MOOCs: The mediating effects of immersive experience and psychological needs satisfaction. *Telematics and Informatics*, 39, 75–91.

Gao, Q., Zheng, H., Sun, R., & Lu, S. (2022). Parent-adolescent relationships, peer relationships, and adolescent mobile phone addiction: The mediating role of psychological needs satisfaction. *Addictive Behaviors*, 129, Article 107260.<https://doi.org/10.1016/j.addbeh.2022.107260>

Gentile, D. A. (2009). Pathological video-game use among youth ages 8 to 18: A national study. *Psychological Science*, 20(5), 594–602.

Gillis, A., & Roskam, I. (2019). Daily exhaustion and support in parenting: Impact on the quality of the parent-child relationship. *Journal of Child and Family Studies*, 28(7), 2007–2016.

Granic, I., Lobel, A., & Engels, R. C. M. E. (2014). The benefits of playing video games. *American Psychologist*, 69(1), 66–78.

Green, C. S., & Bavelier, D. (2003). Action video game modifies visual selective attention. *Nature*, 423(6939), 534–537.

Hamre, R., Smith, O. R. F., Samdal, O., & Haug, E. (2022). Gaming behaviors and the association with sleep duration, social jetlag, and difficulties falling asleep among Norwegian adolescents. *International Journal of Environmental Research and Public Health*, 19(3), Article 1765. <https://doi.org/10.3390/ijerph19031765>

Hsieh, Y.-P., Shen, A. C.-T., Wei, H.-S., Feng, J.-Y., Huang, S. C.-Y., & Hwa, H.-L. (2018). Internet addiction: A closer look at multidimensional parenting practices and child mental health. *Cyberpsychology Behavior and Social Networking*, 21(12), 768–773.

Karabanova, O. A., & Poskrebsheva, N. N. (2013). Adolescent autonomy in parent-child relations. *Procedia-Social and Behavioral Sciences*, 86, 621–628.

Kardefelt-Winther, D. (2014). A conceptual and methodological critique of Internet addiction research: Towards a model of compensatory Internet use. *Computers in Human Behavior*, 31, 351–354.

Ko, C.-H., Yen, J.-Y., Chen, C.-C., Chen, S.-H., & Yen, C.-F. (2005). Gender differences and related factors affecting online gaming addiction among Taiwanese adolescents. *The Journal of Nervous and Mental Disease*, 193(4), 273–277.

Kuhl, J. (1984). Volitional aspects of achievement motivation and learned helplessness: Toward a comprehensive theory of action control. *Progress in Experimental Personality Research*, 13, 99–171.

Lam, L. T., Peng, Z., Mai, J., & Jing, J. (2009). Factors associated with Internet addiction among adolescents. *CyberPsychology & Behavior*, 12(5), 551–555.

Li, D., Zhang, W., Li, X., Zhen, S., & Wang, Y. (2010). Stressful life events and problematic Internet use by adolescent females and males: A mediated moderation model. *Computers in Human Behavior*, 26(5), 1199–1207.

Li, D., Zhou, Y., Li, X., & Zhou, Z. (2016). Perceived school climate and adolescent Internet addiction: The mediating role of deviant peer

affiliation and the moderating role of effortful control. *Computers in Human Behavior*, 60, 54–61.

Li, J., Zhan, D., Zhou, Y., & Gao, X. (2021). Loneliness and problematic mobile phone use among adolescents during the COVID-19 pandemic: The roles of escape motivation and self-control. *Addictive Behaviors*, 118, Article 106857. <https://doi.org/10.1016/j.addbeh.2021.106857>

McDaniel, B. T., & Radesky, J. S. (2018). Technoference: Parent distraction with technology and associations with child behavior problems. *Child Development*, 89(1), 100–109.

Mikolajczak, M., Brianda, M. E., Avalosse, H., & Roskam, I. (2018). Consequences of parental burnout: Its specific effect on child neglect and violence. *Child Abuse & Neglect*, 80, 134–145.

Mikolajczak, M., Gross, J. J., Stinghamber, F., Norberg, A. L., & Roskam, I. (2020). Is parental burnout distinct from job burnout and depressive symptoms? *Clinical Psychological Science*, 8(4), 673–689.

Pettit, G. S., & Bates, J. E. (1989). Family interaction patterns and children's behavior problems from infancy to 4 years. *Developmental Psychology*, 25(3), 413–420.

Przybylski, A. K. (2014). Electronic gaming and psychosocial adjustment. *Pediatrics*, 134(3), e716–e722.

Roskam, I., Brianda, M.-E., & Mikolajczak, M. (2018). A step forward in the conceptualization and measurement of parental burnout: The parental burnout assessment (PBA). *Frontiers in Psychology*, 9, Article 758. <https://doi.org/10.3389/fpsyg.2018.00758>

Roskam, I., Raes, M.-E., & Mikolajczak, M. (2017). Exhausted parents: Development and preliminary validation of the parental burnout inventory. *Frontiers in Psychology*, 8, Article 163. <https://doi.org/10.3389/fpsyg.2017.00163>

Rothbart, M. K., & Bates, J. E. (2006). Temperament. In N. Eisenberg, W. Damon, & R. M. Lerner (Eds.), *Handbook of child psychology: Vol. 3. Social, emotional, and personality development* (6th ed. pp. 99–166). John Wiley & Sons.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78.

Schumacher, P., & Morahan-Martin, J. (2001). Gender, Internet and computer attitudes and experiences. *Computers in Human Behavior*, 17(1), 95–110.

Stavropoulos, V., Alexandraki, K., & Motti-Stefanidi, F. (2013). Flow and telepresence contributing to Internet abuse: Differences according to gender and age. *Computers in Human Behavior*, 29(5), 1941–1948.

Sim, T. N. (2000). Adolescent psychosocial competence: The importance and role of regard for parents. *Journal of Research on Adolescence*, 10(1), 49–64.

Suler, J. R. (1999). To get what you need: Healthy and pathological Internet use. *CyberPsychology & Behavior*, 2(5), 385–393.

Sun, R., Gao, Q., Xiang, Y., Chen, T., Liu, T., & Chen, Q. (2020). Parent-child relationships and mobile phone addiction tendency among Chinese adolescents: The mediating role of psychological needs satisfaction and the moderating role of peer relationships. *Children and Youth Services Review*, 116, Article 105113. <https://doi.org/10.1016/j.childyouth.2020.105113>

Tereshchenko, S., Kasparov, E., Smolnikova, M., Shubina, M., Gorbacheva, N., & Moskalenko, O. (2021). Internet addiction and sleep problems among russian adolescents: A field school-based study. *International Journal of Environmental Research and Public Health*, 18(19), Article 10397. <https://doi.org/10.3390/ijerph181910397>

Toker, S., & Baturay, M. H. (2016). Antecedents and consequences of game addiction. *Computers in Human Behavior*, 55, 668–679.

Varchetta, M., Frascchetti, A., Mari, E., & Giannini, A. M. (2020). Social media addiction, fear of missing out (FoMO) and online vulnerability in university students. *Revista Digital de Investigación En Docencia Universitaria*, 14(1), Article e1187. <https://doi.org/10.19083/ridu.2020.1187>

Wang, M., Qu, X., Chen, X., Wu, X., & Wang, J. (2023). Harsh fathering longitudinally predicts adolescents' Internet gaming addiction via

unfulfilled psychological need. *Youth & Society*, 56(2),0044118X2311686. <https://doi.org/10.1177/0044118X23116866>

Xu, S., Liu, Z., Tian, S., Ma, Z., Jia, C., & Sun, G. (2021). Physical activity and resilience among college students: The mediating effects of basic psychological needs. *International Journal of Environmental Research and Public Health*, 18(7), Article 3722.<https://doi.org/10.3390/ijerph18073722>

Xu, Z., Turel, O., & Yuan, Y. (2012). Online game addiction among adolescents: Motivation and prevention factors. *European Journal of Information Systems*, 21(3), 321–340.

Yang, B., Chen, B.-B., Qu, Y., & Zhu, Y. (2021). Impacts of parental burnout on Chinese youth's mental health: The Role of parents' autonomy support and emotion regulation. *Journal of Youth and Adolescence*, 50(8), 1679–1692.

Yuan, Y., Wang, W., Song, T., & Li, Y. (2022). The mechanisms of parental burnout affecting adolescents' problem behavior. *International Journal of Environmental Research and Public Health*, 19(22), Article 15139.<https://doi.org/10.3390/ijerph192215139>

Chen, W., Li, D., Bao, Z., Yan, Y., & Zhou, Z. (2015). The Impact of parent-child attachment on adolescent problematic Internet use: A moderated mediation model. *Acta Psychologica Sinica*, 47(5), 611–623.

Chen, Y., Zhong, S., Dai, L., Deng, Y., & Liu, X.(2019). Attachment anxiety and social networking sites addiction in college students: A moderated mediating model. *Chinese Journal of Clinical Psychology*, 27(3), 497–500+505.

Cheng, H., Liu, X., Li, Y., & Li, Y. (2021). Is parenting a happy experience? Review on parental burnout. *Psychological Development and Education*, 37(1),146–152.

Gao, W., & Chen, Z. (2006). A study on psychopathology and psychotherapy of Internet addiction. *Advances in Psychological Science*, 14(4), 596–603.

Lin, Y., Liu, Q., Yu, S., & Zhou, Z. (2021). The relationship between parents neglect and online gaming addiction among adolescents: The

mediating role of hope and gender difference. *Psychological Development and Education*, 37(1), 109–119.

Luo, J., Dong, H., Ding, Q., & Li, D. (2017). Cumulative ecological risk and adolescent Internet addiction: a Moderating role of effortful control. *Chinese Journal of Clinical Psychology*, 25(5), 893–896+901.

Ma, N., Zhang, W., Yu, C., Zhu, J., Jiang, Y., & Wu, T. (2017). Perceived school climate and Internet gaming disorder in junior school students: A moderated mediation model. *Chinese Journal of Clinical Psychology*, 25(1), 65–69+74.

Peng, Y., & Zhou, S. (2007). Relationship of Internet addiction and family environment and parental rearing patterns in adolescents. *Chinese Journal of Clinical Psychology*, 15(4), 418–419+439.

Wang, X., Gao, S., & Liu, S. (2022). Impact of social exclusion on college students' Internet addiction: Mediation of self-concept clarity. *China Journal of Health Psychology*, 30(6), 934–939.

Wang, Y., Li, D., Sun, W., Zhao, L., Lai, X., & Zhou, Y. (2017). Parent-child attachment and prosocial behavior among junior high school students: Moderated mediation effect. *Acta Psychologica Sinica*, 49(5), 663–679.

Wen, Z., & Ye, B. (2014). Different methods for testing moderated mediation models: Competitors or backups? *Acta Psychologica Sinica*, 46(5), 714–726.

Wu, S., Zhou, Z., Niu, G., & Liu, M. (2015). Video game transfer phenomena: Type, manifestation and mechanism. *Advances in Psychological Science*, 23(6), 1041–1051.

Xia, F., & Ye, B. (2014). The effect of stressful life events on adolescents' tobacco and alcohol use: The chain mediating effect of basic psychological needs and coping style. *Journal of Psychological Science*, 37(6), 1385–1391.

Ye, B., Yu, S., & Hu, Z. (2013). The effect of stress, gratitude and satisfaction of basic psychological needs on drug use among reform school students. *Psychological Development and Education*, 29(4), 415–423.

Yin, X., & Xian, G. (2021). Effects of parental psychological distress on adolescents' Internet gaming addiction and its path. *Journal of Educational Theory and Practice*, 13(3), 39–45.

Yu, J., Wei, S., Dong, J., Xu, Y., & Liang, Y. (2023). Relationship between harsh parenting and adolescents' Internet gaming disorder: The chain mediation roles of basic psychological needs satisfaction and deviant peer affiliation. *China Journal of Health Psychology*, 31(7), 1090–1095.

Zhang, Y., & Jin, Y. (2019). An overview of action control theory. *Psychological Monthly*, 14(14), 34.

China Internet Network Information Center. (2022). The 2021 national report on the internet usage of minors.2022-03-06. <https://www.cnnic.cn/n4/2022/1201/c135-10691.html>

Zhou, H., & Long, L. (2004). Statistical remedies for common method biases. *Advances in Psychological Science*, 12(6), 942–950.

Zhu, Y., & Jiang, Z. (2022). Parents' phubbing and problematic Internet use in junior high school students: Chain mediation of parent-child cohesion and relatedness need satisfaction. *Chinese Journal of Clinical Psychology*, 30(2), 434–438+487.

养育倦怠与青少年网络游戏成瘾： 基本心理需要满足、意志控制及性别的作用*

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摘要:为探讨养育倦怠对青少年网络游戏成瘾的影响及其作用机制,本研究采用基本心理需要满足量表、网络游戏成瘾量表以及青少年早期气质问卷对青少年进行调查,采用养育倦怠量表对青少年的父亲和母亲进行调查,最终获得2669组家庭的配对数据。在控制了青少年感知的主观家庭社会经济地位后,结果表明:(1)养育倦怠能够显著正向预测青少年网络游戏成瘾;(2)养育倦怠通过降低青少年联结需要满足、自主需要满足和能力需要满足,进而增加其网络游戏成瘾的水平,在中介效应差异比较中,仅自主需要满足的中介效应显著大于能力需要满足的中介效应;(3)“养育倦怠→基本心理需要满足(联结、自主、能力)→网络游戏成瘾”这三组中介效应的后半段均受到青少年意志控制和性别的调节,即对低意志控制和男生个体来说,联结需要满足、自主需要满足和能力需要满足对网络游戏成瘾的影响更强。研究结果可以为青少年网络游戏成瘾的预防和干预提供参考。

关键词:养育倦怠;网络游戏成瘾;基本心理需要满足;意志控制;青少年

分类号:B844

1 引言

中国互联网络信息中心(2022)发布的《2021年全国未成年人互联网使用情况研究报告》显示,我国经常在网上玩游戏的未成年人比例高达62.3%。适当的网络游戏可以满足个体娱乐和社交需要,促进社会化以及认知和情绪能力的发展(Granic et al., 2014; Green & Bavelier, 2003; Przybylski, 2014; 吴四兰等, 2015),但过多的网络游戏会带来成瘾的潜在风险。网络游戏成瘾(Internet gaming addiction)是指个体难以控制地、过度地或强迫性地使用网络游戏并造成身心问题或功能损伤(马娜等, 2017)。青少年是网络游戏成瘾的易感群体(Li et al., 2016),网络游戏成瘾会对青少年的身心健康和学业发展等多方面产生消极影响,例如更严重的睡眠问题、更高的敌意、更多的抑郁情绪、更差的学业表现等(Borgonovi, 2016; Brunborg et al., 2014; Chiu et al., 2004; Hamre et al., 2022;

Tereshchenko et al., 2021)。因此,探讨青少年网络游戏成瘾的成因对促进其健康发展有重要的现实意义。

根据生态系统理论,家庭是个体成长的微环境,父母作为子女的重要他人,对他们的心理和社会能力发展有重要影响(Bronfenbrenner & Ceci, 1994)。已有研究指出,父母缺少温情,存在心理困扰、忽视教养等情况可能导致青少年网络游戏成瘾(林悦等, 2021; 彭阳, 周世杰, 2007; 尹霞云, 先桂瑶, 2021);针对网络成瘾青少年的研究也发现了他们对家庭和父母的不满程度显著高于其他同龄人(Lam et al., 2009)。因此,需要关注父母养育方面因素对青少年网络游戏成瘾的潜在影响。Chen等人(2022)和Yuan等人(2022)的研究结果均表明,养育倦怠是青少年诸多问题行为的重要预测因素,但已有研究尚没有探讨养育倦怠和青少年网络游戏成瘾之间的关系。因此,本研究在以往研究的基础上,聚焦养育倦怠对青少年网络游戏成瘾的影响机

* 基金项目:国家社会科学基金“十三五”规划2020年度教育学青年课题“新时代中小学家校合作有效性评价体系构建与提升策略研究”(CHA200266);北京师范大学中国基础教育质量监测协同创新中心研究生自主课题(BJZK-2021A1-20004)。

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制,包括养育倦怠“如何”影响青少年网络游戏成瘾的中介机制,以及“在何种条件下”养育倦怠对青少年的消极影响被加剧或削弱的调节机制。对于上述问题的深入探讨,有助于为预防和减缓青少年网络游戏成瘾提供理论指导和实证支持。

1.1 养育倦怠与青少年网络游戏成瘾的关系

养育倦怠(parental burnout)是一种父母长期暴露于慢性育儿压力下引发的心理综合征(Roskam et al., 2017; 程华斌等, 2021),具体表现为长期卷入养育活动导致的情感耗竭、失去为人父母的乐趣、有意与子女保持情感距离、怀疑自己能否像以前一样做好父母角色(Roskam et al., 2018)。首先,养育倦怠的父母在养育情境中面临情感耗竭,很难保持良好的状态去养育子女。而青春期正是青少年情感体验的敏感期,当他们感受不到来自父母的情感温暖时,很可能通过网络游戏来满足自身情感需要,导致网络游戏成瘾。Eisenberg等人(2001)的研究表明,父母在亲子互动过程中的消极情绪表达会影响孩子的社会理解能力,进而导致更多行为问题的发生。其次,养育倦怠的父母对自身养育角色感到厌倦、开始有意与子女保持情感距离、养育效能感下降,这将限制父母的主动养育水平。以往研究表明,主动养育水平较低的父母可能疏于对孩子行为的预先指导和管教、更少对孩子表达爱意,导致孩子各种行为问题的发生(Pettit & Bates, 1989)。此外,养育倦怠的父母甚至会表现出忽视、暴力等消极教养行为(Mikolajczak et al., 2018),而青少年可能通过沉溺网络虚拟世界来缓解父母消极养育对自身的不良影响(Chapman et al., 2006)。实证研究表明,经历父母忽视或粗暴养育的青少年更可能出现网络游戏成瘾(林悦等, 2021; 于浚泉等, 2023)。综上,本研究提出假设1,养育倦怠能够显著正向预测青少年网络游戏成瘾,即养育倦怠越严重,青少年网络游戏成瘾水平越高。

1.2 基本心理需要满足在养育倦怠和青少年网络游戏成瘾中的作用

进一步探究养育倦怠对青少年网络游戏成瘾的具体影响机制,能为预防和减缓青少年网络游戏成瘾的发生提供科学依据。基本心理需要满足是个体行为的内在动机(Ryan & Deci, 2000),包括自主需要满足(希望能够自主决定自己的行为)、联结需要满足(希望得到周围人的关爱和支持)和能力需要满足(希望能够胜任一定的任务)三个维度。当心理需要得到满足,个体通常会朝着自我完善、自我提

高的积极方向发展,反之则可能出现心理和行为问题。已有研究表明,基本心理需要满足在父母心理困扰对青少年网络游戏成瘾的影响中起中介作用(尹霞云, 先桂瑶, 2021)。据此推测,基本心理需要满足可能中介了养育倦怠与青少年网络游戏成瘾之间的关系。

具体而言,养育倦怠会导致亲子关系受损(Gillis & Roskam, 2019),当青少年无法感受到与父母之间的密切联系,会破坏青少年的联结需要满足(Gao et al., 2022);积极的亲子关系是青少年自主感的源泉,较差的亲子关系也将威胁青少年的自主需要满足(Karabanova & Poskrebysheva, 2013);消极的亲子关系还可能阻碍青少年自我能力的发展,影响能力需要满足(Sim, 2000)。“需要—满足”理论认为,心理需要被压抑、忽视、转移或停留在表面满足或间接满足,可能导致个体成瘾(Suler, 1999)。网络游戏允许玩家之间线上交流,为青少年提供了联结需要满足;玩家可以自由选择角色、购买装备、发动技能等,为青少年提供了自主需要满足;与其他玩家竞技以达到更高的游戏水平,又为青少年提供了能力需要满足(Wang et al., 2023)。综上,本研究提出假设2,基本心理需要满足的三个维度在养育倦怠和青少年网络游戏成瘾间均起中介作用。

然而,不同维度基本心理需要满足的作用程度可能并不完全一致(Xu et al., 2021)。亲子互动是建立和维持亲子情感联系的主要方式,养育倦怠的父母越来越少地参与互动过程,这更可能导致青少年联结需要满足的缺失。网络游戏又具有社交属性,更吸引缺乏联结需要满足的青少年在网络游戏寻求社会支持。综上,本研究提出假设3,联结需要满足在养育倦怠和青少年网络游戏成瘾中的中介作用更大。

1.3 意志控制的调节作用及性别差异

意志控制(effortful control)是一种个体主动抑制优势反应,激活次优势反应的自我调节能力(Rothbart & Bates, 2006)。意志行动控制理论(Theory of Action Control)认为,动机和意志共同决定了个体的行为(Kuhl, 1984)。其中,动机帮助个体根据自身需要选择预期价值最大的目的行为,意志则帮助个体克服干扰以执行相应的目的行为(张亚辉, 靳媛, 2019)。高意志控制能力是个体健康发展的保护因素,如Li等人(2010)指出高意志控制的青少年通常较少出现问题性互联网使用。具有较高水平意志控制能力的青少年在面临负性生活事件

时也更可能主动采取认知重构、问题解决等适应性方式以应对最初的被动反应倾向 (Carver & Connor-Smith, 2010)。罗金晶等人 (2017) 指出, 意志控制能够缓冲累积生态风险对青少年网络成瘾的负面影响。由此推测, 高意志控制能力的青少年在不同的基本心理需求没有得到满足而通过网络游戏寻求补偿的过程中, 他们可能会重新评估游戏带来的风险和收益, 抑制内心对游戏的过度渴望以避免成瘾。相反, 低意志控制能力的青少年可能期望及时行乐, 在不同的基本心理需求没有得到满足的情况下, 更容易被网络游戏带来的积极情绪体验吸引, 导致成瘾。以往研究也发现, 自我控制调节了逃避动机和问题性手机使用之间的关联, 即自我控制能力越差, 青少年更可能因逃避动机而导致问题性手机使用 (Li et al., 2021)。因此, 本研究提出假设 4, 青少年意志控制能够调节基本心理需要的三个维度对网络游戏成瘾的影响。

性别也是青少年网络游戏成瘾的重要影响因素, 相关研究结果表明, 男生的网络游戏成瘾率高于同龄女生 (Stavropoulos et al., 2013; Toker & Baturay, 2016; 林悦等, 2021)。这可能是因为男生玩网络游戏的经验比女生更多, 动机更强 (Ko et al., 2005)。此外, Stavropoulos 等人 (2013) 指出, 男生在网络游戏中感受到的沉浸式体验高于女生。更强的沉浸感又满足了个体多样化的基本心理需要 (Fang et al., 2019), 可能导致男生更高的网络游戏成瘾风险。因此, 本研究提出假设 5, 基本心理需要满足的三个维度对青少年网络游戏成瘾的影响存在性别差异。

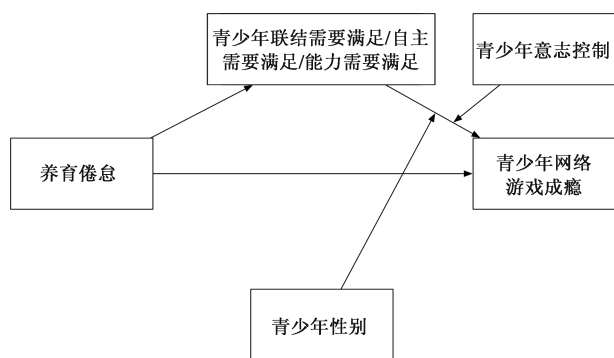


图 1 拟检验的假设模型

2 研究方法

2.1 被试

本研究采用整群抽样的方法, 以北京市和海南省 13 所中学的初一年级学生及其父母为研究

对象。主试为参加过统一培训的心理学研究生。在填答开始前, 主试会征询所有被试参与研究的意愿, 并告知被试可以有选择地对感兴趣的问卷进行填答以及随时退出的权利。本研究共回收 3274 组家庭的数据, 并采取了较为严格的数据纳入标准。家庭任一成员在单变量作答上缺失超过 30% 及无效作答的 605 组家庭的数据被剔除, 最终得到 2669 组有效数据, 有效率 81.52%。2669 名有效学生被试中, 男生 1418 人, 53.10%; 女生 1251 人, 46.90%。在后续研究中, 男生被试编码为 0, 女生被试编码为 1。

2.2 工具

2.2.1 养育倦怠

采用 Roskam 等人 (2018) 编制的养育倦怠量表 (Parental Burnout Assessment), 中文版由国内学者 Cheng 等人 (2020) 翻译。该量表由 23 个题目组成, 包括父母角色的耗竭感、与以往父母角色的自我对比、父母角色的厌烦感以及与子女的情感疏远四个维度。该量表采用 7 级计分, 范围从 0“从不”到 6“每天”。父亲和母亲单独施测, 计算平均分, 得分越高, 表示养育倦怠越严重。本研究中父亲和母亲养育倦怠量表的 Cronbach's α 系数均为 0.98。将父亲和母亲养育倦怠得分取平均值, 合成养育倦怠。

2.2.2 基本心理需要满足

采用 Chen 等人 (2015) 编制的基本心理需要满足量表 (Basic Psychological Need Satisfaction and Frustration Scale)。该量表由 24 个题目组成, 包括三个维度, 分别是联结需要满足、自主需要满足和能力需要满足。该量表采用 5 级计分, 范围从 1“完全不符合”到 5“完全符合”。对反向题反向计分, 计算所有题目的平均分, 得分越高, 表示青少年的基本心理需要满足程度越高。本研究中联结需要满足分量表的 Cronbach's α 系数为 0.82、自主需要满足分量表的 Cronbach's α 系数为 0.73、能力需要满足分量表的 Cronbach's α 系数为 0.84。

2.2.3 网络游戏成瘾

采用 Gentile (2009) 根据 DSM-IV 赌博成瘾量表 (Criteria for Pathological Gambling) 改编的网络游戏成瘾量表。该量表要求被试报告网络游戏成瘾相关症状出现的频率, 共 11 个题目。该问卷采用 3 级计分, 1 为“从不”, 2 为“有时”, 3 为“经常”。将得分进行重新赋值, 选“从不”的不计分, 选“有时”的计 0.5 分, 选“经常”的计 1 分, 计算总分, 分数越高

表示个体的网络游戏成瘾水平越高。本研究中该量表的 Cronbach's α 系数为 0.88。

2.2.4 意志控制

采用青少年早期气质问卷 (Early Adolescent Temperament Questionnaire) 中激活控制、注意和抑制控制三个子维度测量意志控制 (Capaldi & Rothbart, 1992)。问卷共由 15 个题目组成,采用 5 级计分,范围从 1“非常不符合”到 5“非常符合”。对反向题反向计分,计算所有题目的平均分,得分越高表明意志控制能力越高。本研究中该量表的 Cronbach's α 系数为 0.82。

2.2.5 控制变量

结合以往研究结果(王晓娟 等, 2022),本研究将主观家庭社会经济地位作为控制变量纳入所有分析。主观家庭社会经济地位由十个阶梯组成的 MacArthur 量表来测量 (Adler et al., 2000)。阶梯越高则得分越高,也就意味着青少年感知到家庭成员拥有更高的教育水平和社会地位,家庭生活品质更好。

2.3 数据处理方法

本研究使用 SPSS 26.0 对数据进行描述性统计分析和相关分析。在此基础上采用 Process 宏程序进行有调节的中介效应分析,并采用 Bootstrap 法获得中介效应参数估计的标准误及置信区间,设定重复抽样次数为 5000,通过估算的 95% 置信区间判断参数的显著性。

3 结果

3.1 共同方法偏差检验

除养育倦怠由父母分别报告外,其他变量都由

青少年报告,故青少年报告的结果可能存在共同方法偏差。本研究采用 Harman 单因素法对青少年报告的结果进行检验,共提取出 8 个主成分,其中一个因子解释的总方差变异量为 25.46%,小于周浩和龙立荣(2004)提出的 40% 临界标准,故共同方法偏差对本研究结果的影响不大。

3.2 各变量之间的相关

对养育倦怠、联结需要满足、自主需要满足、能力需要满足、意志控制、网络游戏成瘾、青少年性别和青少年感知的主观家庭社会经济地位进行相关分析。结果表明,养育倦怠和主观家庭社会经济地位、基本心理需要满足的三个维度、意志控制呈显著负相关,和网络游戏成瘾呈显著正相关。基本心理需要满足的三个维度均和主观家庭社会经济地位、意志控制呈显著正相关,均和网络游戏成瘾呈显著负相关。意志控制和主观家庭社会经济地位呈显著正相关,和网络游戏成瘾呈显著负相关。网络游戏成瘾和主观家庭社会经济地位呈显著负相关。此外,青少年性别与联结需要满足、能力需要满足、意志控制和网络游戏成瘾相关显著,具体表现为,女生的联结需要满足和意志控制水平更高,男生的能力需要满足和网络游戏成瘾水平更高。详见表 1。

3.3 养育倦怠与青少年网络游戏成瘾:基本心理需要满足、意志控制及性别的作用

本文各研究变量间的方差膨胀因子最大为 2.69,小于临界值 4,不存在严重的多重共线问题。对除性别外的其他变量均进行标准化处理,根据温忠麟和叶宝娟(2014)的建议,分步检验含有调节的中介效应。

表 1 各研究变量之间的描述性统计结果和相关分析

	1	2	3	4	5	6	7	8
1 性别	1							
2 主观家庭社会经济地位	—	1						
3 养育倦怠	-0.01	-0.13 ***	1					
4 联结需要满足	0.05 *	0.26 ***	-0.20 ***	1				
5 自主需要满足	0.03	0.22 ***	-0.17 ***	0.71 ***	1			
6 能力需要满足	-0.09 **	0.27 ***	-0.14 ***	0.68 ***	0.71 ***	1		
7 意志控制	0.05 **	0.27 ***	-0.22 ***	0.64 ***	0.65 ***	0.64 ***	1	
8 网络游戏成瘾	-0.23 ***	-0.15 ***	0.17 ***	-0.33 ***	-0.36 ***	-0.33 ***	-0.49 ***	1
$M \pm SD_{\text{全体}}$	—	5.96 $\pm$ 1.56	1.71 $\pm$ 0.97	3.75 $\pm$ 0.76	3.53 $\pm$ 0.68	3.46 $\pm$ 0.80	3.65 $\pm$ 0.60	2.11 $\pm$ 2.08
$M \pm SD_{\text{男生}}$	—	5.98 $\pm$ 1.57	1.72 $\pm$ 0.96	3.71 $\pm$ 0.73	3.51 $\pm$ 0.66	3.53 $\pm$ 0.77	3.62 $\pm$ 0.61	2.56 $\pm$ 2.21
$M \pm SD_{\text{女生}}$	—	5.95 $\pm$ 1.54	1.70 $\pm$ 0.97	3.79 $\pm$ 0.79	3.54 $\pm$ 0.70	3.38 $\pm$ 0.84	3.68 $\pm$ 0.59	1.61 $\pm$ 1.81

注: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$,下同。

表 2 青少年基本心理需要满足三个维度的中介作用及差异

变量	β	95% bootLLCI	95% bootULCI
联结需要满足	0.015 *	0.006	0.025
自主需要满足	0.027 *	0.018	0.038
能力需要满足	0.012 *	0.006	0.019
联结需要满足 v. s. 自主需要满足	-0.012	-0.023	0.003
联结需要满足 v. s. 能力需要满足	0.003	-0.009	0.015
自主需要满足 v. s. 能力需要满足	0.015 *	0.003	0.029

第一步建立一个中介模型,采用 Process 中的模型 4 检验基本心理需要满足的三个维度^①是否在养育倦怠和青少年网络游戏成瘾之间起中介作用,并进一步考查了中介作用差异,在所有分析中均控制了主观家庭社会经济地位。结果表明,养育倦怠影响青少年网络游戏成瘾的总效应和直接效应均显著($\beta = 0.15, p < 0.001$; $\beta = 0.10, p < 0.001$),假设 1 成立。养育倦怠对基本心理需要满足三个维度的预测作用均显著($\beta = -0.17, p < 0.05$; $\beta = -0.15, p < 0.001$; $\beta = -0.10, p < 0.001$)。基本心理需要满足的三个维度对青少年网络游戏成瘾的预测作用均显著($\beta = -0.09, p < 0.05$; $\beta = -0.19, p < 0.001$; $\beta = -0.12, p < 0.001$)。中介分析显示,基本心理需要满足的三个维度在养育倦怠对青少年网络游戏成瘾的影响中均起中介作用,中介效应分别占比 9.66%、17.63%、7.58%,假设 2 成立。进一步的差异检验表明,自主需要满足的中介效应显著大于能力需要满足的中介效应,而与联结需要满足的中介效应差异不显著,联结需要满足的中介效应和能力需要满足的中介效应差异也不显著,假设 3 不成立。详见表 2。

第二步分别建立三个有调节的中介模型,采用 Process 中的模型 16 分别检验养育倦怠通过联结需要满足、自主需要满足、能力需要满足影响青少年网络游戏成瘾中的中介效应是否受到青少年意志控制和性别的调节。结果表明:三个模型中,养育倦怠对基本心理需要满足三个维度的预测作用均显著;基本心理需要满足的三个维度对青少年网络游戏成瘾的预测作用均显著;青少年意志控制对青少年网络游戏成瘾的预测作用均显著;基本心理需要满足的三个维度和意志控制的交互作用对青少年网络游戏成瘾的预测作用均显著,说明意志控制均调节了基

本心理需要满足的三个维度与网络游戏成瘾的关系,故假设 4 成立;性别对青少年网络游戏成瘾的预测作用均显著;基本心理需要满足的三个维度和性别的交互作用对网络游戏成瘾的预测作用均显著,说明在基本心理需要满足的三个维度与网络游戏成瘾的关系中均存在性别差异,故假设 5 成立。详见表 3。

为了更好地理解调节效应的作用趋势,分别进行简单斜率检验。结果表明,意志控制能力较差($M - 1SD$)的男生,基本心理需要满足的三个维度对网络游戏成瘾的预测作用均显著($\beta = -0.14, p < 0.001$; $\beta = -0.23, p < 0.001$; $\beta = -0.25, p < 0.001$)。意志控制能力较高($M + 1SD$)的男生,联结需要满足对网络游戏成瘾的预测作用不显著($\beta = -0.01, p > 0.05$),自主需要满足和能力需要满足对网络游戏成瘾的预测作用仍显著($\beta = -0.07, p < 0.05$; $\beta = -0.09, p < 0.01$),但预测程度均低于意志控制能力较差的男生。意志控制能力较差($M - 1SD$)的女生,联结需要满足对网络游戏成瘾的预测作用不显著($\beta = -0.05, p > 0.05$),自主需要满足和能力需要满足对网络游戏成瘾的预测作用显著($\beta = -0.12, p < 0.001$; $\beta = -0.12, p < 0.001$)。意志控制能力较高($M + 1SD$)的女生,联结需要满足显著正向预测网络游戏成瘾($\beta = 0.08, p < 0.01$),自主需要满足和能力需要满足对网络游戏成瘾的预测作用则不显著($\beta = 0.04, p > 0.05$; $\beta = 0.05, p > 0.05$)。详见图 2。

进一步检验被青少年意志控制和性别调节下的中介效应,结果发现,基本心理需要满足三个维度的中介效应均受到意志控制和性别的调节。详见表 4。

① 在后文中按联结需要满足、自主需要满足和能力需要满足的顺序呈现结果。

表3 意志控制和性别的调节作用检验结果

变量	方程1					方程2				
	β	t	95% LLCI	95% ULCI	R^2	β	t	95% LLCI	95% ULCI	R^2
因变量：联结需要满足					0.09	因变量：网络游戏成瘾				
主观家庭社会经济地位	0.24 ***	12.72	0.20	0.27		-0.02	-0.92	-0.05	0.02	
养育倦怠	-0.17 ***	-9.00	-0.20	-0.13		0.06 ***	3.71	0.03	0.10	
联结需要满足						-0.08 **	-2.78	-0.13	-0.02	
意志控制						-0.45 ***	-21.04	-0.50	-0.41	
联结需要满足 × 意志控制						0.07 ***	4.14	0.03	0.10	
性别						-0.40 ***	-12.29	-0.47	-0.34	
联结需要满足 × 性别						0.10 **	2.94	0.03	0.16	
因变量：自主需要满足					0.07	因变量：网络游戏成瘾				0.30
主观家庭社会经济地位	0.20 ***	10.40	0.16	0.23		-0.02	-0.10	-0.05	0.02	
养育倦怠	-0.15 ***	-7.75	-0.18	-0.11		0.06 ***	3.61	0.03	0.09	
自主需要满足						-0.15 ***	-5.40	-0.21	-0.10	
意志控制						-0.42 ***	-19.02	-0.46	-0.38	
自主需要满足 × 意志控制						0.08 ***	5.07	0.05	0.11	
性别						-0.40 ***	-12.39	-0.47	-0.34	
自主需要满足 × 性别						0.11 ***	3.47	0.05	0.18	
因变量：能力需要满足					0.08	因变量：网络游戏成瘾				0.30
主观家庭社会经济地位	0.25 ***	13.55	0.22	0.29		-0.01	-0.59	-0.04	0.02	
养育倦怠	-0.10 ***	-5.44	-0.14	-0.07		0.07 ***	3.93	0.03	0.10	
能力需要满足						-0.17 ***	-5.97	-0.23	-0.11	
意志控制						-0.41 ***	-18.79	-0.45	-0.37	
能力需要满足 × 意志控制						0.08 ***	5.15	0.05	0.11	
性别						-0.43 ***	-12.89	-0.49	-0.36	
能力需要满足 × 性别						0.13 ***	4.12	0.07	0.20	

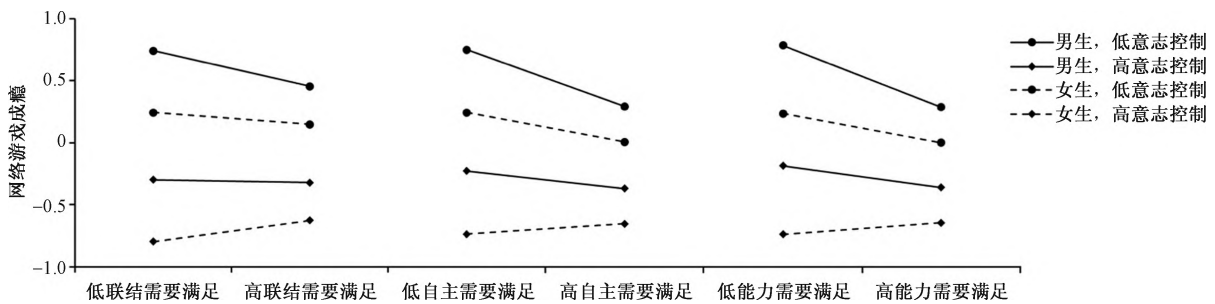


图2 青少年意志控制和性别在基本心理需要满足的三个维度与网络游戏成瘾关系中的调节作用

表4 基本心理需要满足三个维度被青少年意志控制和性别调节的中介效应差异

中介变量	调节变量	中介效应差异	95% bootLLCI	95% bootULCI
联结需要满足	意志控制	0.011 *	0.005	0.018
	性别	0.016 *	0.005	0.028
自主需要满足	意志控制	0.012 *	0.007	0.018
	性别	0.017 *	0.008	0.027
能力需要满足	意志控制	0.008 *	0.005	0.013
	性别	0.014 *	0.007	0.022

4 讨论

本研究深入研讨养育倦怠对青少年网络游戏成瘾的影响及具体的内在机制,并进一步分析了基本心理需要满足三个不同维度的中介作用差异。结果表明,养育倦怠通过基本心理需要满足的三个维度

进而影响网络游戏成瘾,且三个不同维度基本心理需要满足的中介作用存在差异。研究还发现,在联结、自主和能力需要没有得到满足时,低意志控制能力和男生个体的网络游戏成瘾水平更高。

4.1 养育倦怠对青少年网络游戏成瘾的影响

本研究首次揭示了养育倦怠对青少年网络游戏成瘾的影响,拓展了对养育倦怠影响子女身心健康发展的认识。与养育压力领域的研究结果类似,父母高水平的养育压力与青少年更严重的网络成瘾有关(Bozoglan & Kumar, 2022)。养育倦怠会导致青少年网络游戏成瘾,导致这种结果的可能原因有:(1)养育倦怠作为一种家庭风险因素,会威胁亲子间的关系质量(Gillis & Roskam, 2019),如果青少年感到亲子间的情感联系变得薄弱,会错把游戏中的

奖赏机制和积极情绪体验当做心灵的慰藉(陈武等, 2015), 更有可能网络成瘾(Hsieh et al., 2018)。(2)养育倦怠的父母减少养育投入, 使得他们对子女的异常行为的感知能力减弱(McDaniel & Radesky, 2018), 父母监管缺位而导致青少年沉迷网络游戏(Xu et al., 2012)。

4.2 基本心理需要满足在养育倦怠和青少年网络游戏成瘾中的作用

本研究发现, 父母养育倦怠水平越高, 青少年的联结需要、自主需要和能力需要满足水平越低, 进而增加网络游戏成瘾风险, 这与“需要—满足”理论的解释相符(Suler, 1999)。养育倦怠的父母会通过减少对孩子的情感付出、仅提供工具性支持, 避免心理资源过度消耗, 从而实现他们自身倦怠状态的缓解(Mikolajczak et al., 2020)。这可能导致父母忽略青少年寻求心理需要满足的信号, 不利于亲子关系的健康发展, 威胁青少年基本心理需要的满足(朱颖, 姜兆萍, 2022)。个体在基本心理需要未得到满足时, 可能会寻求替代性满足(Ryan & Deci, 2000)。游戏不但具有匿名性、便利性的特点, 而且青少年能够从中体验到掌握感和成就感, 这使得游戏成为他们寻求替代性满足的重要方式(Kardefelt-Winther, 2014)。现实世界中基本心理需要的匮乏与虚拟世界中基本心理需要的满足之间落差越大, 个体网络游戏成瘾的风险越高(Allen & Anderson, 2018)。Bender 和 Gentile(2020)的研究结果也得出了相似的结论, 即游戏中的高需求满足程度促使个体成瘾。物质成瘾领域的相关研究也指出, 当青少年的基本心理需要没有获得满足时, 烟酒、毒品成瘾等行为明显增加(叶宝娟等, 2013; 夏扉, 叶宝娟, 2014)。

需要指出的是, 联结需要满足的中介作用并不是最大的, 这与假设并不一致。可能是因为随着儿童进入青春期, 他们更多通过与同伴交往来满足自身的联结需要(Gao et al., 2022)。相反, 从亲子互动中获得的联结需要满足则随年龄增加逐渐减少。但本研究仍发现基本心理需要满足三个维度的中介作用差异, 即自主需要满足的中介作用显著大于能力需要满足的中介作用。这可能是因为倦怠的父母自主支持教养水平较低(Yang et al., 2021), 青少年的自主需要没有得到充分满足。而建立自我同一性是青春期个体面临的主要发展任务(Erikson, 1956), 他们更渴望优先获得自主感以满足自身发展需要。因此, 与其他需要满足相比, 养育倦怠可能

会更多地通过影响青少年的自主需要满足, 进而增加网络游戏成瘾的风险。

4.3 意志控制的调节作用及性别差异

本研究在意志行动控制理论的视角下, 考查了在不同意志控制水平的青少年中, 不同维度的基本心理需要满足影响网络游戏成瘾的个体差异。结果表明, 意志控制在青少年基本心理需要的三个维度没有得到满足时能起到“雪中送炭”的作用, 这与王艳辉等人(2017)的研究结果相似。具体表现为随着青少年意志控制能力的提升, 基本心理需要满足的三个维度对青少年网络游戏成瘾的预测作用逐渐下降, 在青少年意志控制能力处于高水平时, 基本心理需要满足的三个维度在养育倦怠和青少年网络游戏成瘾之间的中介作用被阻断。这一现象可以由“失补偿”假说解释(高文斌, 陈祉妍, 2006), 该假说认为高意志控制的青少年更容易形成“建设性补偿”, 即适度游戏; 低意志控制的青少年则容易出现“病理性补偿”, 即网络游戏成瘾。换言之, 高意志控制能力的青少年可能具有较好的自我行为管理能力, 能够较好地把握使用网络游戏来满足自身基本心理需要的“度”, 而低意志控制能力的青少年可能一味地沉迷, 难以自拔。

本研究还考查了不同维度的基本心理需要满足和网络游戏成瘾关系中的性别差异。虽然男女生网络游戏成瘾均受基本心理需要满足的三个维度影响, 但是在男生中影响更甚。这可能是因为男生对互联网技术的态度通常更为积极和投入(Schumacher & Morahan-Martin, 2001), 男生玩网络游戏的经验更丰富(Ko et al., 2005); 另一方面, 男生倾向于在网上寻找主导活动或内容, 那些特别依赖挑战、控制的交互式网络游戏更吸引男生参与(Chou et al., 2005), 同时男生也从中能获得更强的沉浸体验(Stavropoulos et al., 2013)。网络游戏的沉浸式体验更能满足男生的基本心理需要, 进一步导致了更高的网络游戏成瘾风险。前人研究也发现了类似模式, 性别显著调节了归属需要对社交网络成瘾的影响(陈云祥等, 2019), 本研究将这种性别差异拓展到不同维度的基本心理需要满足和网络游戏成瘾之间的关系。

需要指出的是, 高意志控制能力的女生, 联结需要满足的程度越高, 游戏成瘾风险反而增加。这可能有两方面因素的共同作用; 其一, 意志控制在调节联结需要满足和网络游戏成瘾的关系中, 对低意志控制能力的个体来说, 网络游戏成瘾水平整体较高,

这可能掩盖了联结需要满足对网络游戏成瘾的影响。对高意志控制能力的个体来说,网络游戏成瘾水平整体较低,此时联结需要满足对网络游戏成瘾的影响更明显;其二,女生往往表现出更高水平的社交媒体成瘾和关系需求(Varchetta et al., 2020),女生可能倾向于在网络游戏中寻求联结需要的满足,进一步导致网络成瘾。综上,要警惕高意志控制能力的女生因联结需要的过度满足反而导致网络游戏成瘾风险增加的可能。

4.4 研究意义

与以往单独探讨整体基本心理需要满足的中介作用或者意志控制的调节作用研究不同,本研究不仅探讨了不同维度基本心理需要满足的作用机制,还将“需要—满足”理论和意志行动控制理论整合在一个模型框架中,这有助于更好地理解不同维度的基本心理需要满足和意志控制在驱动个体行为时如何共同作用。本研究将性别作为预测变量而非控制变量,发现了不同维度的基本心理需要满足影响网络游戏成瘾过程中的性别差异。此外,同时对青少年意志控制和性别的调节作用加以探讨,能够进一步明确中介作用起效及效应大小的不同条件,这些发现将为预防和减缓青少年网络游戏成瘾带来一些启示。首先,青少年时期是个体身心发展的疾风骤雨期,青少年内心企盼独立的种子萌芽,但又难以真正脱离父母。此时,来自父母的情感支持和行为约束仍然非常重要,这将有效防止青少年误入歧途。其次,关注并提高青少年基本心理需要满足的水平,特别是自主需要满足的水平,能够有效降低他们对网络游戏的依赖;注重培养青少年意志控制能力,有助于降低他们在网络游戏世界中寻求心理需要满足而导致成瘾的潜在风险。再次,男生,尤其是低意志控制的男生,面临因基本心理需要没有得到满足而导致网络游戏成瘾的风险更高,应予以特别关注。最后,当父母意识到自身养育资源不足以应对养育需求时,应当及时寻求外界帮助,提升青少年健康福祉。

4.5 研究不足与展望

本研究仍存在一些局限,可在未来的研究中进一步完善。首先,受限于横断设计,不能做出因果推断。因此,有必要在未来的研究中采用纵向设计对本研究得出的结论进行验证和扩展。其次,青少年网络游戏成瘾在本研究中作为结果变量,但值得注意的是,青少年网络游戏成瘾也可能导致父母养育效能降低而产生倦怠情绪。未来研究可以在纵向设

计的基础上考察两者间的相互关系。再次,在高意志控制能力的女生中,联结需要满足对网络游戏成瘾的特殊作用有待于在更广泛的年龄段中进行检验。最后,除了父母以外,同伴是青少年青春期最主要的重要他人(Brown et al., 1993),积极的同伴关系可以为他们提供充足的情感支持,满足其基本心理需要,降低网络游戏成瘾风险(Sun et al., 2020)。未来的研究可以进一步考虑同伴支持的缓冲作用,以期多途径保护青少年免受养育倦怠的影响。

5 结论

综上所述,本研究获得如下主要结论:(1)养育倦怠能够显著正向预测青少年网络游戏成瘾;(2)养育倦怠通过损害青少年联结、自主和能力需要满足,进而影响网络游戏成瘾,在中介效应差异比较中,仅发现自主需要满足的中介效应显著大于能力需要满足的中介效应;(3)相比于高意志控制和女生个体,联结、自主、能力需要满足这三组中介效应均在低意志控制和男生个体中更强。

参考文献:

- Adler, N. E., Epel, E. S., Castellazzo, G., & Ickovics, J. R. (2000). Relationship of subjective and objective social status with psychological and physiological functioning: Preliminary data in healthy white women. *Health Psychology, 19*(6), 586–592.
- Allen, J. J., & Anderson, C. A. (2018). Satisfaction and frustration of basic psychological needs in the real world and in video games predict Internet gaming disorder scores and well-being. *Computers in Human Behavior, 84*, 220–229.
- Bender, P. K., & Gentile, D. A. (2020). Internet gaming disorder: Relations between needs satisfaction in-game and in life in general. *Psychology of Popular Media, 9*(2), 266–278.
- Borgonovi, F. (2016). Video gaming and gender differences in digital and printed reading performance among 15-year-olds students in 26 countries. *Journal of Adolescence, 48*(1), 45–61.
- Bozoglan, B., & Kumar, S. (2022). Parenting styles, parenting stress and hours spent online as predictors of child Internet addiction among children with autism. *Journal of Autism and Developmental Disorders, 52*(10), 4375–4383.
- Bronfenbrenner, U., & Ceci, S. J. (1994). Nature-nurture reconceptualized in developmental perspective: A bioecological model. *Psychological Review, 101*(4), 568–586.
- Brown, B. B., Mounts, N., Lamborn, S. D., & Steinberg, L. (1993). Parenting practices and peer group affiliation in adolescence. *Child Development, 64*(2), 467–482.
- Brunborg, G. S., Mentzoni, R. A., & Frøyland, L. R. (2014). Is video gaming, or video game addiction, associated with depression, academic achievement, heavy episodic drinking, or conduct problems?

- Journal of Behavioral Addictions*, 3(1), 27–32.
- Capaldi, D. M., & Rothbart, M. K. (1992). Development and validation of an early adolescent temperament measure. *The Journal of Early Adolescence*, 12(2), 153–173.
- Carver, C. S., & Connor-Smith, J. (2010). Personality and coping. *Annual Review of Psychology*, 61(1), 679–704.
- Chapman, A. L., Gratz, K. L., & Brown, M. Z. (2006). Solving the puzzle of deliberate self-harm: The experiential avoidance model. *Behaviour Research and Therapy*, 44(3), 371–394.
- Chen, B., Vansteenkiste, M., Beyers, W., Boone, L., Deci, E. L., Van der Kaap-Deeder, J., ... Verstuyf, J. (2015). Basic psychological need satisfaction, need frustration, and need strength across four cultures. *Motivation and Emotion*, 39(2), 216–236.
- Chen, B.-B., Qu, Y., Yang, B., & Chen, X. (2022). Chinese mothers' parental burnout and adolescents' internalizing and externalizing problems: The mediating role of maternal hostility. *Developmental Psychology*, 58(4), 768–777.
- Cheng, H., Wang, W., Wang, S., Li, Y., Liu, X., & Li, Y. (2020). Validation of a Chinese version of the parental burnout assessment. *Frontiers in Psychology*, 11, Article 321. <https://doi.org/10.3389/fpsyg.2020.00321>
- Chiu, S. I., Lee, J. Z., & Huang, D. H. (2004). Video game addiction in children and teenagers in Taiwan. *Cyberpsychology & Behavior*, 7(5), 571–581.
- Chou, C., Condrón, L., & Belland, J. C. (2005). A review of the research on Internet addiction. *Educational Psychology Review*, 17(4), 363–388.
- Eisenberg, N., Gershoff, E. T., Fabes, R. A., Shepard, S. A., Cumberland, A. J., Losoya, S. H., ... Murphy, B. C. (2001). Mother's emotional expressivity and children's behavior problems and social competence: Mediation through children's regulation. *Developmental Psychology*, 37(4), 475–490.
- Erikson, E. H. (1956). The problem of ego identity. *Journal of the American Psychoanalytic Association*, 4(1), 56–121.
- Fang, J., Tang, L., Yang, J., & Peng, M. (2019). Social interaction in MOOCs: The mediating effects of immersive experience and psychological needs satisfaction. *Telematics and Informatics*, 39, 75–91.
- Gao, Q., Zheng, H., Sun, R., & Lu, S. (2022). Parent-adolescent relationships, peer relationships, and adolescent mobile phone addiction: The mediating role of psychological needs satisfaction. *Addictive Behaviors*, 129, Article 107260. <https://doi.org/10.1016/j.addbeh.2022.107260>
- Gentile, D. A. (2009). Pathological video-game use among youth ages 8 to 18: A national study. *Psychological Science*, 20(5), 594–602.
- Gillis, A., & Roskam, I. (2019). Daily exhaustion and support in parenting: Impact on the quality of the parent-child relationship. *Journal of Child and Family Studies*, 28(7), 2007–2016.
- Granic, I., Lobel, A., & Engels, R. C. M. E. (2014). The benefits of playing video games. *American Psychologist*, 69(1), 66–78.
- Green, C. S., & Bavelier, D. (2003). Action video game modifies visual selective attention. *Nature*, 423(6939), 534–537.
- Hamre, R., Smith, O. R. F., Samdal, O., & Haug, E. (2022). Gaming behaviors and the association with sleep duration, social jet-lag, and difficulties falling asleep among Norwegian adolescents. *International Journal of Environmental Research and Public Health*, 19(3), Article 1765. <https://doi.org/10.3390/ijerph19031765>
- Hsieh, Y.-P., Shen, A. C.-T., Wei, H.-S., Feng, J.-Y., Huang, S. C.-Y., & Hwa, H.-L. (2018). Internet addiction: A closer look at multidimensional parenting practices and child mental health. *Cyberpsychology Behavior and Social Networking*, 21(12), 768–773.
- Karabanova, O. A., & Poskrebsheva, N. N. (2013). Adolescent autonomy in parent-child relations. *Procedia - Social and Behavioral Sciences*, 86, 621–628.
- Kardefelt-Winther, D. (2014). A conceptual and methodological critique of Internet addiction research: Towards a model of compensatory Internet use. *Computers in Human Behavior*, 31, 351–354.
- Ko, C.-H., Yen, J.-Y., Chen, C.-C., Chen, S.-H., & Yen, C.-F. (2005). Gender differences and related factors affecting online gaming addiction among Taiwanese adolescents. *The Journal of Nervous and Mental Disease*, 193(4), 273–277.
- Kuhl, J. (1984). Volitional aspects of achievement motivation and learned helplessness: Toward a comprehensive theory of action control. *Progress in Experimental Personality Research*, 13, 99–171.
- Lam, L. T., Peng, Z., Mai, J., & Jing, J. (2009). Factors associated with Internet addiction among adolescents. *CyberPsychology & Behavior*, 12(5), 551–555.
- Li, D., Zhang, W., Li, X., Zhen, S., & Wang, Y. (2010). Stressful life events and problematic Internet use by adolescent females and males: A mediated moderation model. *Computers in Human Behavior*, 26(5), 1199–1207.
- Li, D., Zhou, Y., Li, X., & Zhou, Z. (2016). Perceived school climate and adolescent Internet addiction: The mediating role of deviant peer affiliation and the moderating role of effortful control. *Computers in Human Behavior*, 60, 54–61.
- Li, J., Zhan, D., Zhou, Y., & Gao, X. (2021). Loneliness and problematic mobile phone use among adolescents during the COVID-19 pandemic: The roles of escape motivation and self-control. *Addictive Behaviors*, 118, Article 106857. <https://doi.org/10.1016/j.addbeh.2021.106857>
- McDaniel, B. T., & Radesky, J. S. (2018). Technoference: Parent distraction with technology and associations with child behavior problems. *Child Development*, 89(1), 100–109.
- Mikolajczak, M., Brianda, M. E., Avalosse, H., & Roskam, I. (2018). Consequences of parental burnout: Its specific effect on child neglect and violence. *Child Abuse & Neglect*, 80, 134–145.
- Mikolajczak, M., Gross, J. J., Stinglhamber, F., Norberg, A. L., & Roskam, I. (2020). Is parental burnout distinct from job burnout and depressive symptoms? *Clinical Psychological Science*, 8(4), 673–689.
- Pettit, G. S., & Bates, J. E. (1989). Family interaction patterns and children's behavior problems from infancy to 4 years. *Developmental*

- Psychology*, 25(3), 413–420.
- Przybylski, A. K. (2014). Electronic gaming and psychosocial adjustment. *Pediatrics*, 134(3), e716–e722.
- Roskam, I., Brianda, M. – E., & Mikolajczak, M. (2018). A step forward in the conceptualization and measurement of parental burnout: The parental burnout assessment (PBA). *Frontiers in Psychology*, 9, Article 758. <https://doi.org/10.3389/fpsyg.2018.00758>
- Roskam, I., Raes, M. – E., & Mikolajczak, M. (2017). Exhausted parents: Development and preliminary validation of the parental burnout inventory. *Frontiers in Psychology*, 8, Article 163. <https://doi.org/10.3389/fpsyg.2017.00163>
- Rothbart, M. K., & Bates, J. E. (2006). Temperament. In N. Eisenberg, W. Damon, & R. M. Lerner (Eds.), *Handbook of child psychology: Vol. 3. Social, emotional, and personality development* (6th ed. pp. 99–166). John Wiley & Sons.
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78.
- Schumacher, P., & Morahan – Martin, J. (2001). Gender, Internet and computer attitudes and experiences. *Computers in Human Behavior*, 17(1), 95–110.
- Stavropoulos, V., Alexandraki, K., & Motti – Stefanidi, F. (2013). Flow and telepresence contributing to Internet abuse: Differences according to gender and age. *Computers in Human Behavior*, 29(5), 1941–1948.
- Sim, T. N. (2000). Adolescent psychosocial competence: The importance and role of regard for parents. *Journal of Research on Adolescence*, 10(1), 49–64.
- Suler, J. R. (1999). To get what you need: Healthy and pathological Internet use. *CyberPsychology & Behavior*, 2(5), 385–393.
- Sun, R., Gao, Q., Xiang, Y., Chen, T., Liu, T., & Chen, Q. (2020). Parent – child relationships and mobile phone addiction tendency among Chinese adolescents: The mediating role of psychological needs satisfaction and the moderating role of peer relationships. *Children and Youth Services Review*, 116, Article 105113. <https://doi.org/10.1016/j.childyouth.2020.105113>
- Tereshchenko, S., Kasparov, E., Smolnikova, M., Shubina, M., Gorbacheva, N., & Moskalenko, O. (2021). Internet addiction and sleep problems among russian adolescents: A field school – based study. *International Journal of Environmental Research and Public Health*, 18(19), Article 10397. <https://doi.org/10.3390/ijerph181910397>
- Toker, S., & Baturay, M. H. (2016). Antecedents and consequences of game addiction. *Computers in Human Behavior*, 55, 668–679.
- Varchetta, M., Frascchetti, A., Mari, E., & Giannini, A. M. (2020). Social media addiction, fear of missing out (FoMO) and online vulnerability in university students. *Revista Digital de Investigación En Docencia Universitaria*, 14(1), Article e1187. <https://doi.org/10.19083/ridu.2020.1187>
- Wang, M., Qu, X., Chen, X., Wu, X., & Wang, J. (2023). Harsh fathering longitudinally predicts adolescents' internet gaming addiction via unfulfilled psychological need. *Youth & Society*, 56(2), 0044118X2311686. <https://doi.org/10.1177/0044118X231168666>
- Xu, S., Liu, Z., Tian, S., Ma, Z., Jia, C., & Sun, G. (2021). Physical activity and resilience among college students: The mediating effects of basic psychological needs. *International Journal of Environmental Research and Public Health*, 18(7), Article 3722. <https://doi.org/10.3390/ijerph18073722>
- Xu, Z., Turel, O., & Yuan, Y. (2012). Online game addiction among adolescents: Motivation and prevention factors. *European Journal of Information Systems*, 21(3), 321–340.
- Yang, B., Chen, B. – B., Qu, Y., & Zhu, Y. (2021). Impacts of parental burnout on Chinese youth's mental health: The Role of parents' autonomy support and emotion regulation. *Journal of Youth and Adolescence*, 50(8), 1679–1692.
- Yuan, Y., Wang, W., Song, T., & Li, Y. (2022). The mechanisms of parental burnout affecting adolescents' problem behavior. *International Journal of Environmental Research and Public Health*, 19(22), Article 15139. <https://doi.org/10.3390/ijerph192215139>
- 陈武, 李董平, 鲍振宙, 闫昱文, 周宗奎. (2015). 亲子依恋与青少年的问题性网络使用: 一个有调节的中介模型. *心理学报*, 47(5), 611–623.
- 陈云祥, 钟姝, 戴莉, 邓衍鹤, 刘翔平. (2019). 大学生依恋焦虑对社交网络成瘾的影响: 有调节的中介模型. *中国临床心理学杂志*, 27(3), 497–500 + 505.
- 程华斌, 刘霞, 李艺敏, 李永鑫. (2021). 养育是一种幸福的体验吗? 养育倦怠述评. *心理发展与教育*, 37(1), 146–152.
- 高文斌, 陈祉妍. (2006). 网络成瘾病理心理机制及综合心理干预研究. *心理科学进展*, 14(4), 596–603.
- 林悦, 刘勤学, 余思, 周宗奎. (2021). 父母忽视与青少年网络游戏成瘾的关系: 希望的中介作用和性别的调节作用. *心理发展与教育*, 37(1), 109–119.
- 罗金晶, 董洪宁, 丁晴雯, 李董平. (2017). 累积生态风险对青少年网络成瘾的影响: 意志控制的调节作用. *中国临床心理学杂志*, 25(5), 893–896 + 901.
- 马娜, 张卫, 喻承甫, 朱键军, 江艳平, 吴涛. (2017). 初中生感知学校氛围与网络游戏障碍: 一个有调节的中介效应模型. *中国临床心理学杂志*, 25(1), 65–69 + 74.
- 彭阳, 周世杰. (2007). 青少年网络成瘾与家庭环境、父母教养方式的关系. *中国临床心理学杂志*, 15(4), 418–419 + 439.
- 王晓娟, 高树青, 刘帅. (2022). 社会排斥对大学生网络成瘾的影响: 自我概念清晰性的中介作用. *中国健康心理学杂志*, 30(6), 934–939.
- 王艳辉, 李董平, 孙文强, 赵力燕, 赖雪芬, 周月月. (2017). 亲子依恋与初中生亲社会行为: 有调节的中介效应. *心理学报*, 49(5), 663–679.
- 温忠麟, 叶宝娟. (2014). 有调节的中介模型检验方法: 竞争还是替补? *心理学报*, 46(5), 714–726.
- 吴四兰, 周宗奎, 牛更枫, 刘美婷. (2015). 视频游戏迁移现象: 类型、表现及作用机制. *心理科学进展*, 23(6), 1041–1051.
- 夏扉, 叶宝娟. (2014). 压力性生活事件对青少年烟酒使用的影响: 基本心理需要和应对方式的链式中介作用. *心理科学*, 37(6), 1385–1391.
- 叶宝娟, 余树英, 胡竹菁. (2013). 压力、感恩和基本心理需要满足对工读生毒品使用的影响机制. *心理发展与教育*, 29(4), 415

- 423.

尹霞云, 先桂瑶. (2021). 父母心理困扰对青少年网络游戏成瘾的影响及其路径. *当代教育理论与实践*, 13(3), 39 - 45.

于浚泉, 魏淑华, 董及美, 徐艳利, 梁艳伟. (2023). 粗暴养育与青少年网络游戏成瘾的关系: 基本心理需要满足和越轨同伴交往的链式中介作用. *中国健康心理学杂志*, 31(7), 1090 - 1095.

张亚辉, 靳媛. (2019). 意志行动控制理论概述. *心理月刊*, 14(14), 34.

中国互联网络信息中心. (2022). 2021 年全国未成年人互联网使用情况研究报告. 2022 - 03 - 06 取自 <https://www.cnnic.cn/n4/2022/1201/c135-10691.html>

周浩, 龙立荣. (2004). 共同方法偏差的统计检验与控制方法. *心理科学进展*, 12(6), 942 - 950.

朱颖, 姜兆萍. (2022). 父母低头行为和初中生问题性网络使用: 亲子亲合和归属需要满足的链式中介. *中国临床心理学杂志*, 30(2), 434 - 438 + 487.

Parental Burnout and Adolescents' Internet Gaming Addiction: Roles of Basic Psychological Need Satisfaction, Effortful Control and Gender

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Abstract: To examine the influence of parental burnout on adolescents' Internet gaming addiction and mechanism, the Basic Psychological Needs Satisfaction Scale, Internet Gaming Addiction Scale and Early Adolescent Temperament Questionnaire were applied to investigate adolescents, and the Parental Burnout Assessment was employed to investigate fathers and mothers. Ultimately, 2669 families' triadic data were collected. After controlling the subjective family socioeconomic status perceived by adolescents, the results showed that: (1) Parental burnout significantly and positively predicted adolescents' Internet gaming addiction; (2) Parental burnout increased the risk of adolescents' Internet gaming addiction by diminishing their need satisfaction for relatedness, autonomy, and competence. When comparing the differences in the mediating effects, only the mediating effect of autonomy need satisfaction was significantly larger than that of competence need satisfaction; (3) The later part of the three mediating effects of "parental burnout → basic psychological need satisfaction (relatedness, autonomy, competence) → Internet gaming addiction" were moderated by adolescents' effortful control and gender. Among lower effortful control and male individuals, relatedness need satisfaction, autonomy need satisfaction, and competence need satisfaction all had stronger impacts on Internet gaming addiction. These findings provided insights for the prevention and intervention of adolescents' Internet gaming addiction.

Key words: parental burnout; Internet gaming addiction; basic psychological need satisfaction; effortful control; adolescents